

Mini Project 2

Folded Cascode Fully Differential Amplifier
With CMFB

Part 1: g_m/ID Design Charts

NMOS Charts:

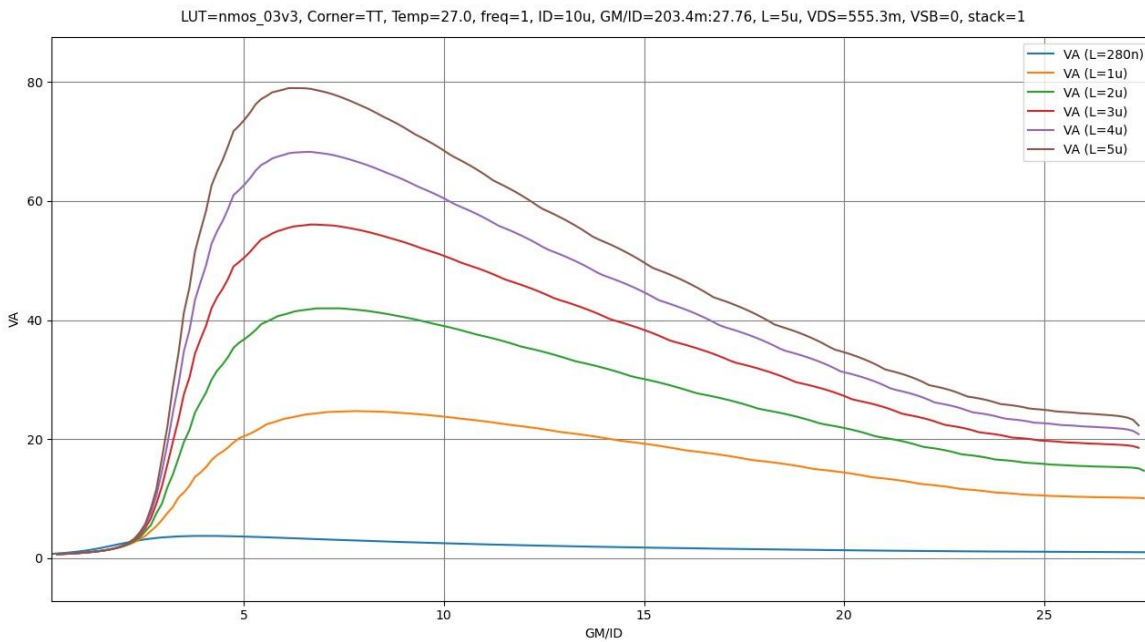


Figure 1 VA vs gm/ID

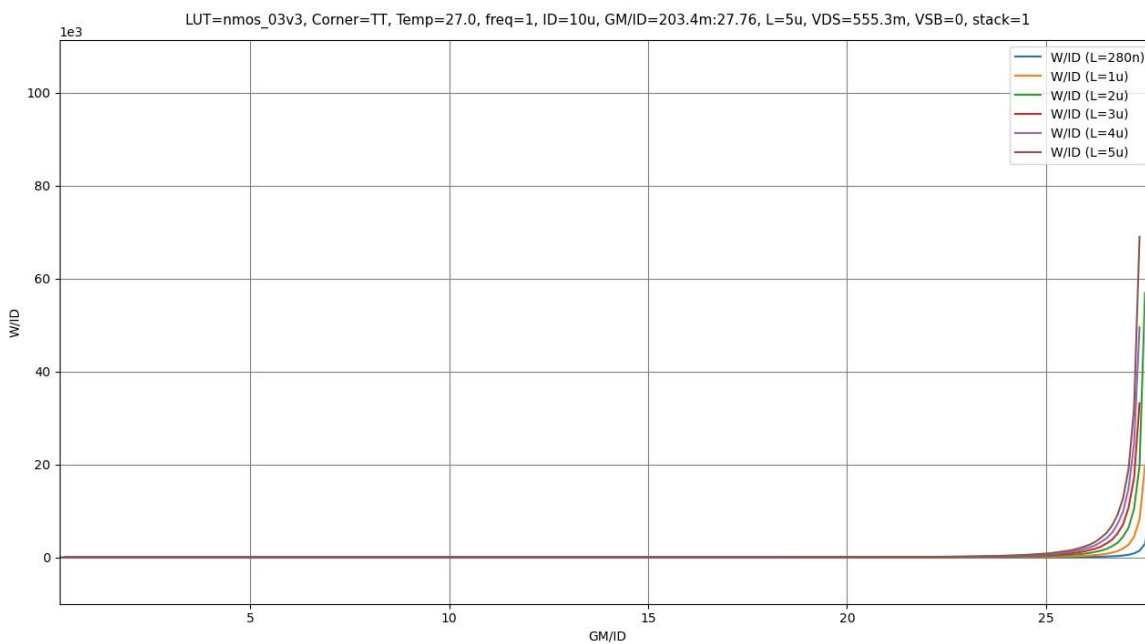


Figure 2 W/ID vs gm/ID

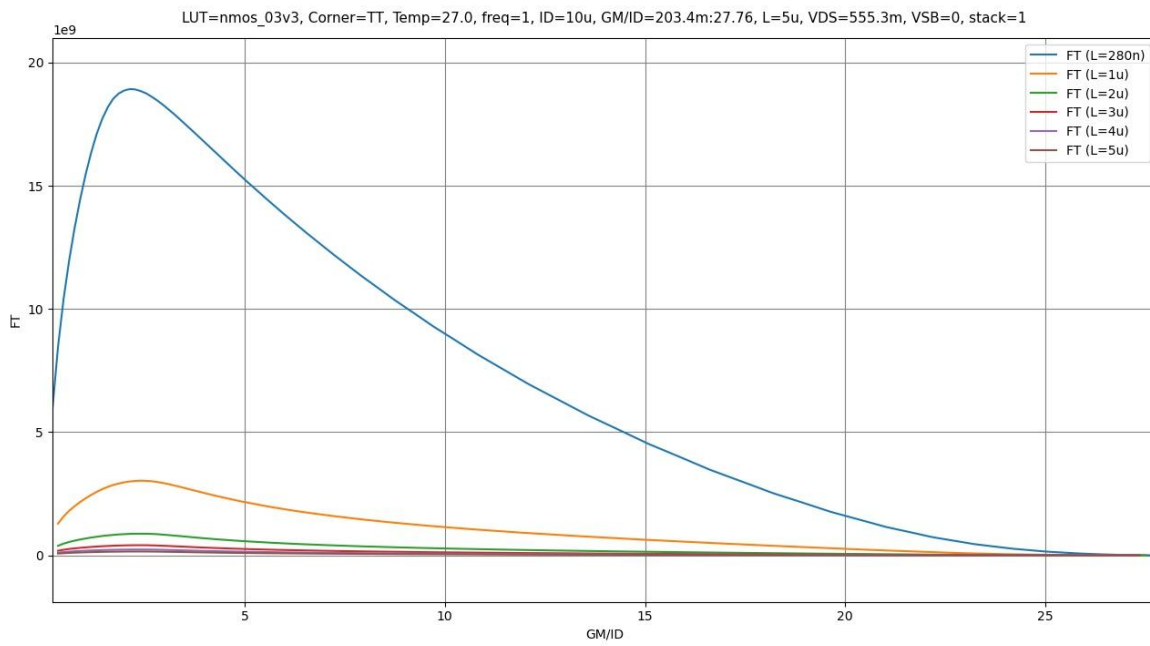


Figure 3 f_t vs g_m/I_D

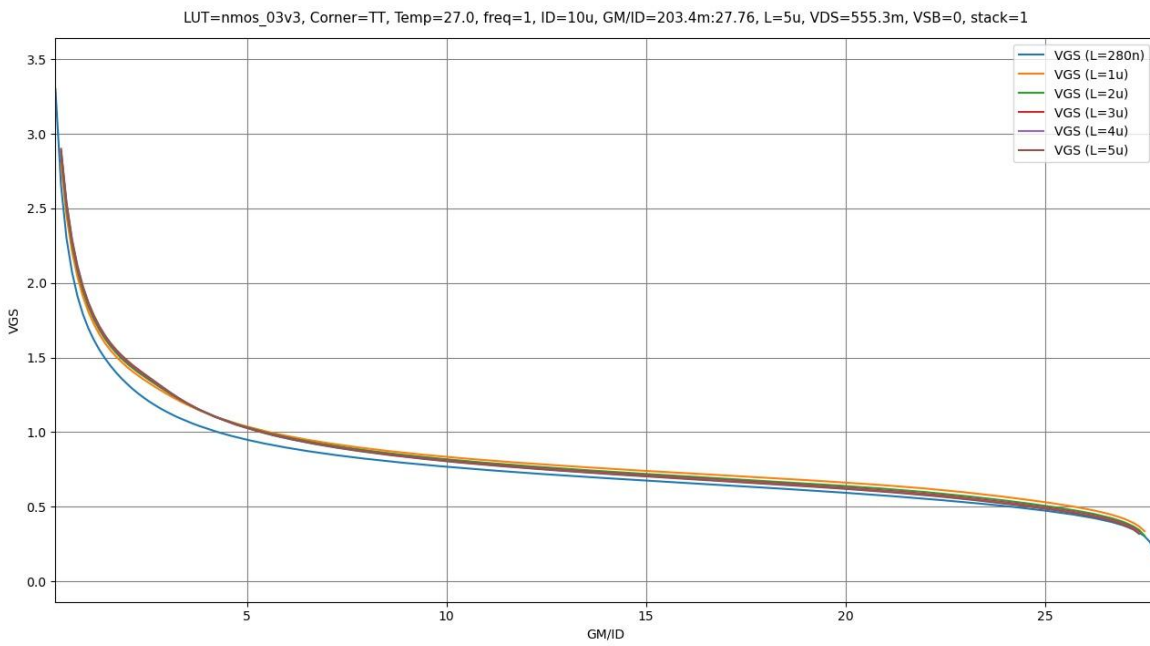


Figure 4 V_{GS} vs g_m/I_D

PMOS Charts:

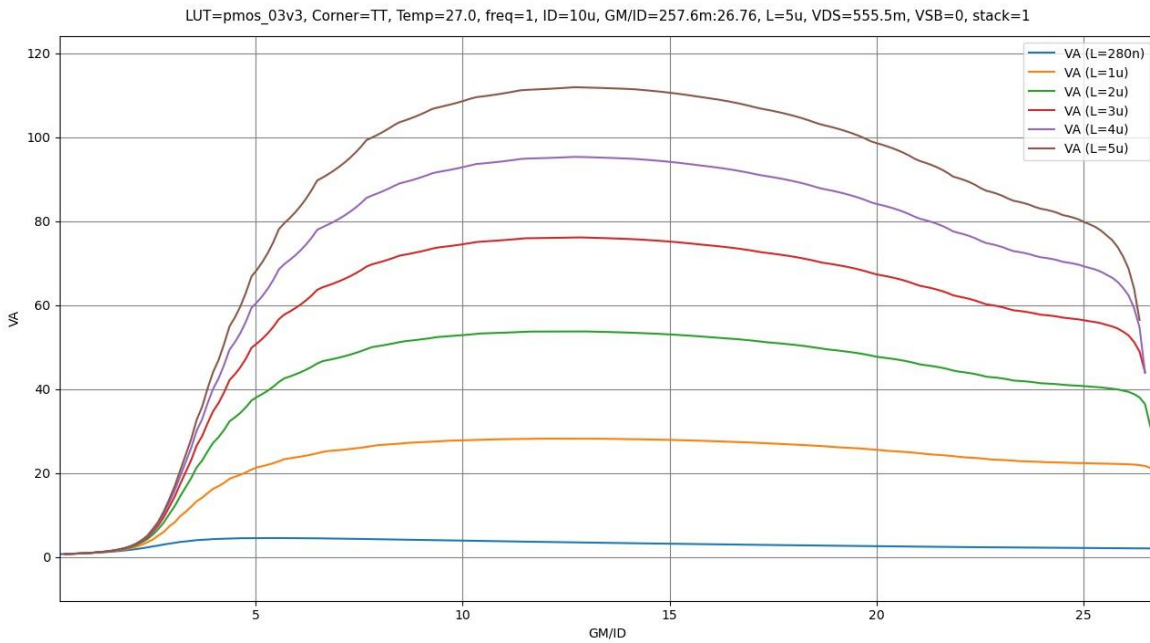


Figure 5 VA vs gm/ID

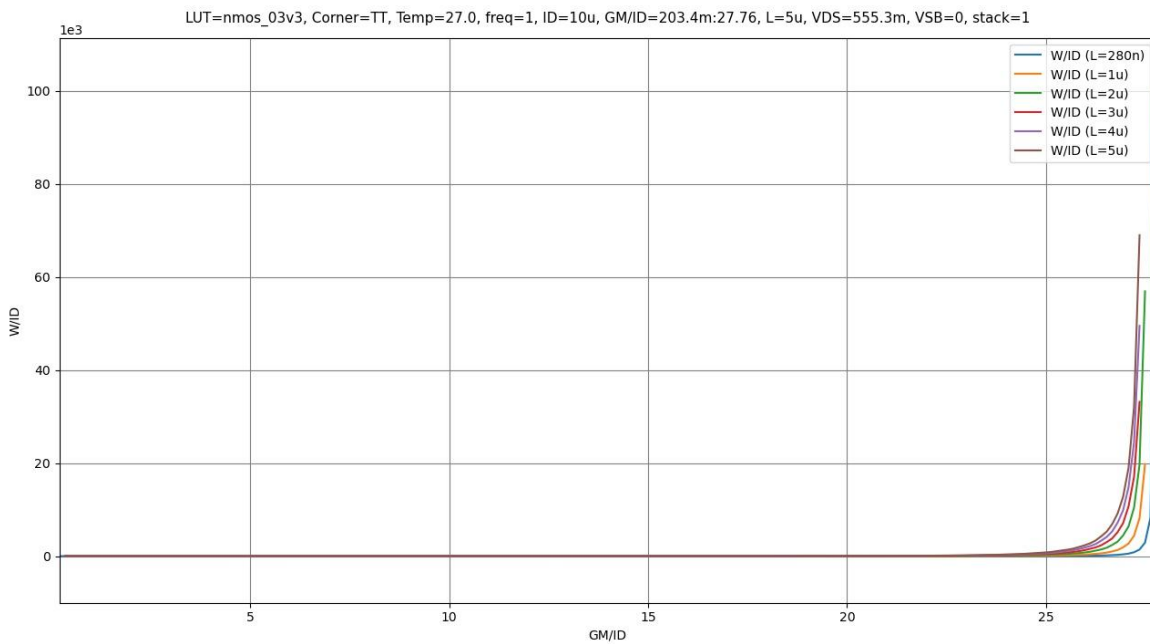


Figure 6 W/ID vs gm/ID

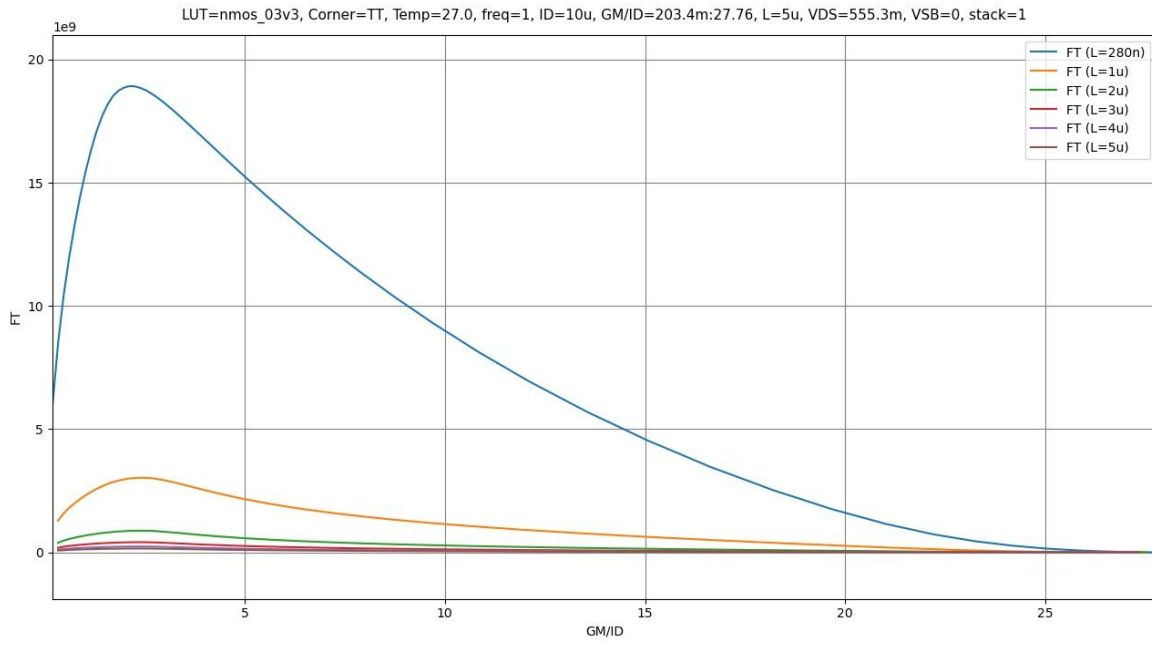


Figure 7 f_t vs g_m/I_D

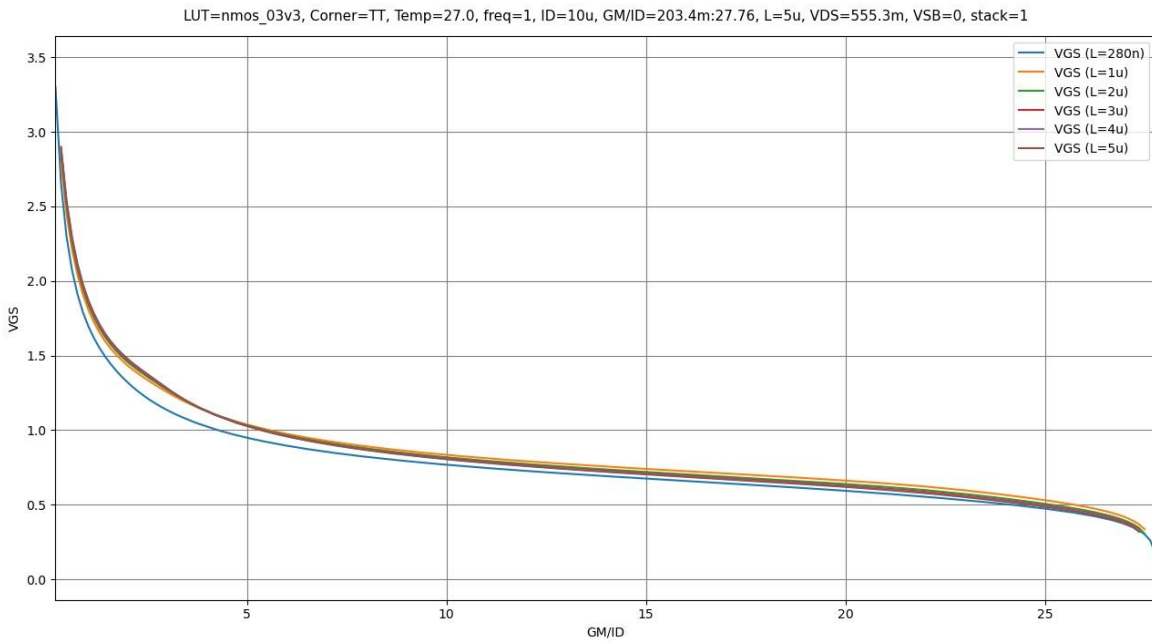


Figure 8 V_{GS} vs g_m/I_D

Part 2: OTA Design

Required Specifications:

Technology	GF180MCU
Supply voltage	2.5V
Closed loop gain	2
Phase margin at the required ACL	$\geq 70^\circ$
CM input range – low	≤ 0
CM input range – high	$\geq 1V$
Differential output swing	1.2Vpk-to-pk
Load	500fF
DC Loop gain	60dB
CL settling time for 1% error	100ns

Design Observations:

Input Range Closer to the ground rail, calls for PMOS Input Stage

Required gain is huge.

Wide Output Swing

Thus, a Folded Cascode Amplifier with PMOS Input stage is the best topology for this spec.

We can take into consideration the loading effect seen by the feedback capacitor for the Closed Loop gain to make our analysis easier later on:

$$\beta = \frac{C_F}{C_F + C_{IN}} = \frac{1p}{1p + 2p} = \frac{1}{3}$$

$$LG = 60dB \rightarrow 1000 = A_{OL} * \beta \rightarrow A_{OL} = \mathbf{3000}$$

We will use these values of gain for every pair of the design.

Input Transistors:

Using the spec for 1% error time settling:

$$e^{-\frac{T}{\tau}} * 100 = 1 \rightarrow \tau = 21.7ns$$

$$BW_{CL} = \frac{1}{2\pi\tau} = 7.33MHz \approx 7.5MHz \text{ (For easier Analysis)}$$

Taking Loading Effect into consideration and Output capacitance as it is comparable to load capacitance:

$$GBW_{CL} = A_{CL} * BW_{CL} = 15MHz = \frac{gm_{in}}{2\pi(C_{out} + C_F + C_L)} \rightarrow gm_{in} \approx 150\mu S$$

Assume $gm/id = 15$ MI or WI

$$I_D = 10\mu A$$

Assume $L = 500nm \rightarrow$ Short L to reduce input capacitances

$$\text{Assume } V_{DS} = \frac{V_{DD}}{3} \approx 0.8333$$

Put into ADT

LUT: pmos_03v3
 Corner: TT [All]
 Temp (°C): 27.0 [All]
 Frequency: 1
 ID: 10u
 gm/ID: 15
 L: 500n
 VDS: 0.8333
 VSB: 0
 Stack: 1

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	500n
4	W	26.32u
5	VGS	848.5m
6	VDS	833.3m
7	VSB	0
8	gm/ID	14.93
9	Vstar	133.9m

Figure 9 Snippet for ADT For Input Pair

$$L = 500nm \quad , \quad W = 26.5\mu m$$

PMOS Current Mirror Load:

Assume Large L: For lower mirroring error

$$L = 1\mu m$$

Assume SI, MI: Lower gm thus Lower Noise

$$\frac{gm}{I_D} = 10$$

$$I_B = 2 * I_D = 10\mu A$$

$$V_{DS} = \frac{V_{DD}}{3} = 0.8333$$

Into ADT:

LUT ?

Corner ☐ All ?

Temp (°C) ☐ All ?

Frequency ?

ID ?

gm/ID ?

L ?

VDS ?

VSB ?

Stack ?

Results:

	Name	TT-27.0
1	ID	20u
2	IG	N/A
3	L	1u
4	W	38.88u
5	VGS	943.5m
6	VDS	833.3m
7	VSB	0
8	gm/ID	9.896
9	Vstar	202.1m

Figure 10 ADT Snippet for Current Mirror Load

$$L = 1\mu m \quad , \quad W = 38.88\mu m \approx 39\mu m$$

NMOS Current Mirror:

Same Assumptions as previously:

Assume Large L: For lower mirroring error

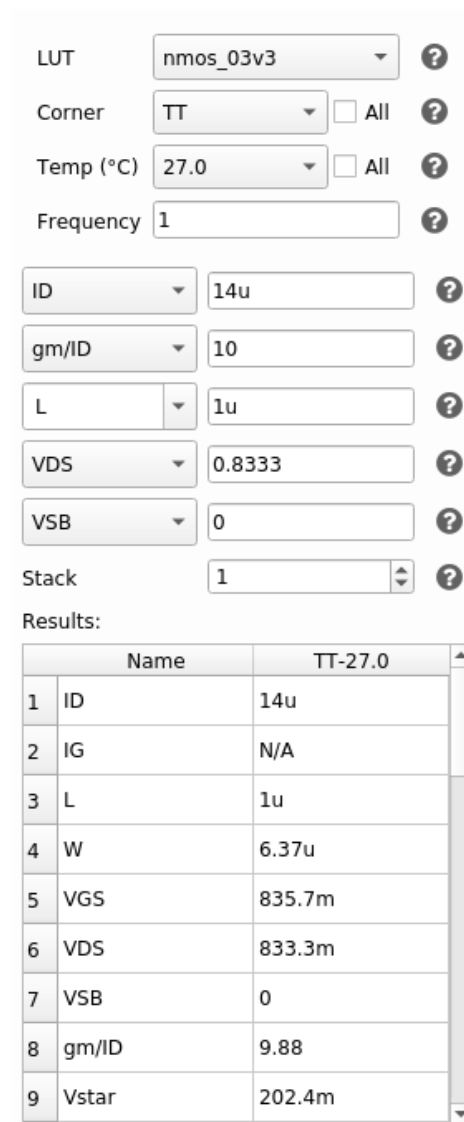
$$L = 1\mu m$$

Assume SI, MI: Lower gm thus Lower Noise

$$\frac{gm}{I_D} = 10$$

$$I_B = 10\mu + 4\mu = 14\mu A \rightarrow \text{Explained in the next section}$$

$$V_{DS} = \frac{V_{DD}}{3} = 0.8333$$



LUT: nmos_03v3

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 14u

gm/ID: 10

L: 1u

VDS: 0.8333

VSB: 0

Stack: 1

Results:

Name	TT-27.0
1 ID	14u
2 IG	N/A
3 L	1u
4 W	6.37u
5 VGS	835.7m
6 VDS	833.3m
7 VSB	0
8 gm/ID	9.88
9 Vstar	202.4m

Figure 11 ADT Snippet for NMOS Current Mirror

$$L = 1\mu m \quad , \quad W = 6.37\mu m$$

Current Mirror:

Sizing the Top current mirror transistor with 1 fifth the Width calculated in the first part to run 2/5 of the current through the CG branch this paper showed that a splitting ratio higher than 1 yields better results, thus I'm using a splitting ratio of 2.5:

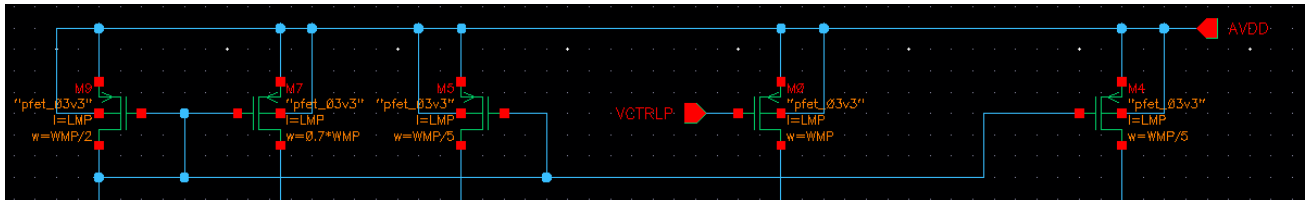


Figure 12 Tail Current Mirror

Paper Link:

https://www.researchgate.net/publication/339758769_Optimum_Split_Ratio_for_Folded_Cascode_OTA_Bias_Current_A_Qualitative_and_Quantitative_Study

Cascode Devices:

Assuming Moderate Length:

$$L = 500nm$$

And WI inversion for higher gain:

$$\frac{gm}{ID} = 15$$

$$V_{DS} = \frac{V_{DD} - \frac{V_{DD}}{3}}{3} = 0.5556V$$

(as third the voltage is already taking by the nmos Current Source)

$$I_D = 4\mu A$$

Into ADT for NMOS Device:

Results:

	Name	TT-27.0
1	ID	4u
2	IG	N/A
3	L	500n
4	W	3u
5	VGS	752.5m
6	VDS	555.6m
7	VSB	0
8	gm/ID	14.9
9	Vstar	134.2m

Figure 13 ADT Snippet for NMOS Cascode Device

$$L = 500nm, W = 3\mu m$$

Into ADT for PMOS Device:

LUT ?
 Corner ☐ All ?
 Temp (°C) ☐ All ?
 Frequency ?
 ID ?
 gm/ID ?
 L ?
 VDS ?
 VSB ?
 itack ?
 results:

	Name	TT-27.0
1	ID	4u
2	IG	N/A
3	L	500n
4	W	10.68u
5	VGS	848.9m
6	VDS	555.6m
7	VSB	0
8	gm/ID	14.95
9	Vstar	133.8m

Figure 14 ADT Snippet for PMOS Cascode Device

$$L = 500\text{nm} \quad , \quad W = 10.68\mu\text{m}$$

Bias Voltages:

$$V_{CASCN} = V_{GScascN} + V_{MN}^* = 0.9525V$$

$$V_{CASCN} = V_{DD} - V_{GScascP} - V_{MP}^* = 1.45V$$

Choosing slighter higher and lower values to ensure transistor operate in saturation:

$$V_{CASCN} = 1V$$

$$V_{CASCN} = 1.3V$$

Initial Design Point:

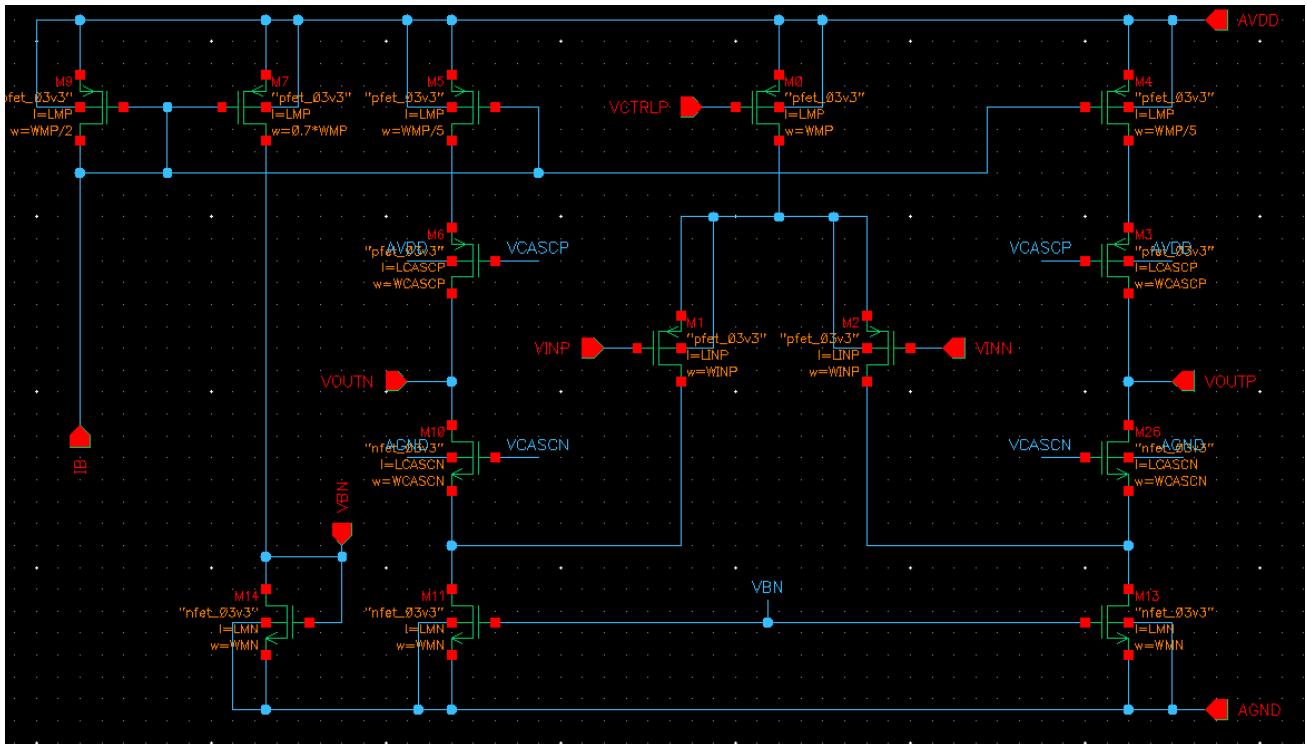


Figure 15 Final Sizing for First Iteration and Topology

Global Variables		
☒	IB	10u
☒	VCASCN	1
☒	VCASCP	1.3
☒	VICM	0.5
☒	VIDIFF	1
☒	VREF	1.25
☒	WCASCP	10.68u
☒	LCASCP	500n
☒	WCASCN	3u
☒	LCASCN	500n
☒	WMN	6.37u
☒	LMN	1u
☒	WMP	39u
☒	LMP	1u
☒	WINP	26.5u
☒	LINP	500n

Click to add variable

Figure 16 Initial Design Parameters

To Test the initial Design point, I set up a simple testbench using the behavioral CMFB Network and tested DC Point and Gain Values to fine tune them.

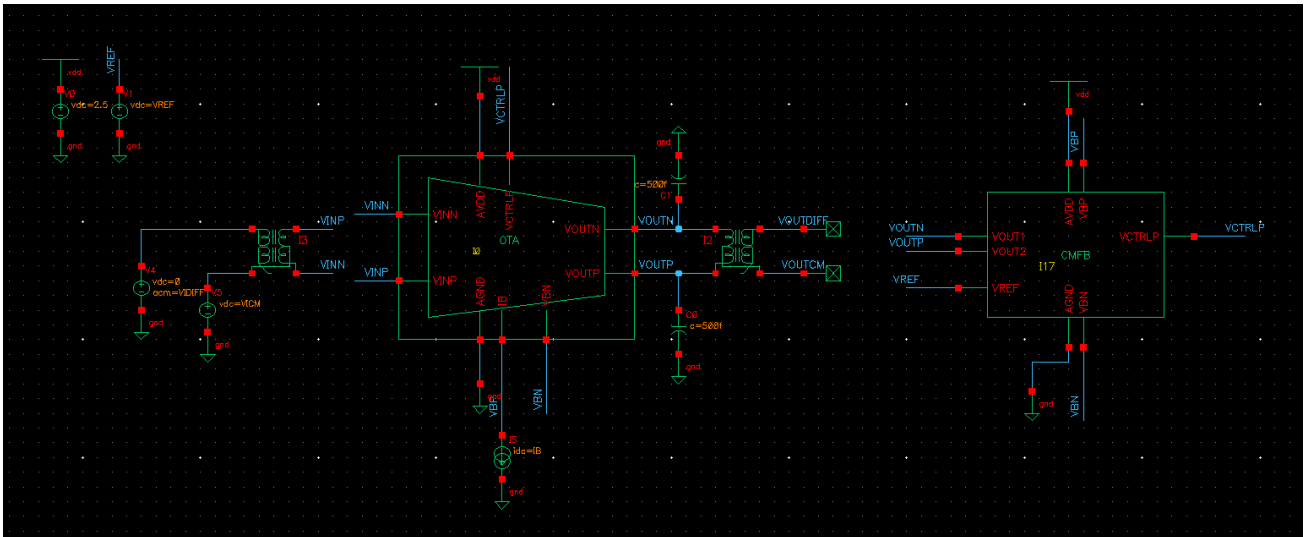


Figure 17 OL Testbench Schematic

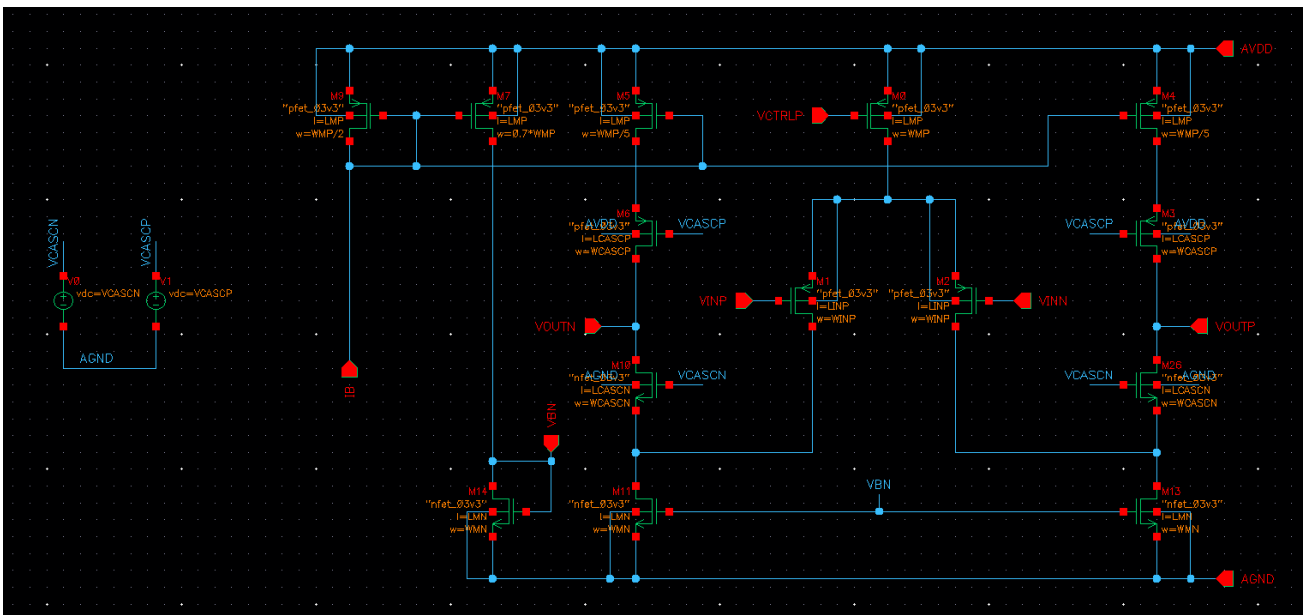


Figure 18 OTA Schematic

I simulated the Bias voltages on the side at the beginning using ideal voltage sources for fine tuning, I will be designing a bias circuit later.

Biasing Circuit:

Using the calculated dimensions, I acquired acceptable results right away, so I moved on directly to the biasing circuit.

I wanted to design the circuit how it would be designed in real life using a practical approach, so I searched the web for biasing circuit for the folded cascode OTA till I found one being discussed in the following forum discussion: [folded cascode biasing | Forum for Electronics](#)

Biasing Circuit with initial Sizing:

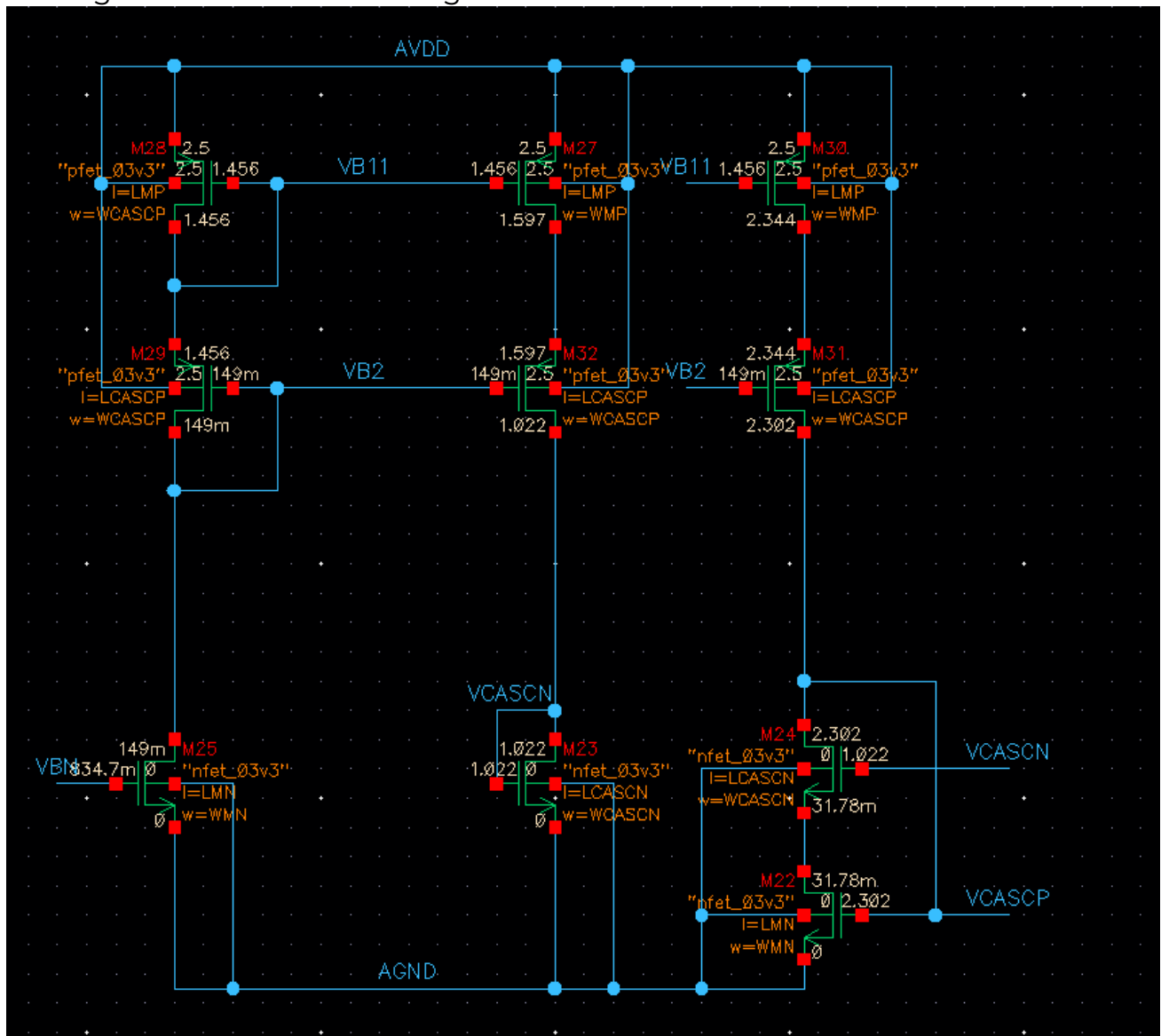


Figure 19 Folded Cascode Biasing Circuit

I made some adjustments to the circuit to better suit my needs:

I used an NMOS transistor connected to the NMOS Tail current mirror from the OTA to drive this circuit

I used the same sizing as the Cascode devices to be biased and the mirror devices they are following

The Cascode configuration in this circuit seems to be for better stability or higher accuracy of the circuit

For sizing I had no specific plan to go about it, so I essentially put the same sizing as the cascode, and mirror devices required to be biased and started adjusting the widths to control VGS and reach the required voltages calculated in the different part

I noticed the three NMOS transistors in the bottom right of the circuit have the higher effect, so I focused my fine tuning on them.

Final Sizing of the Bias Circuit:

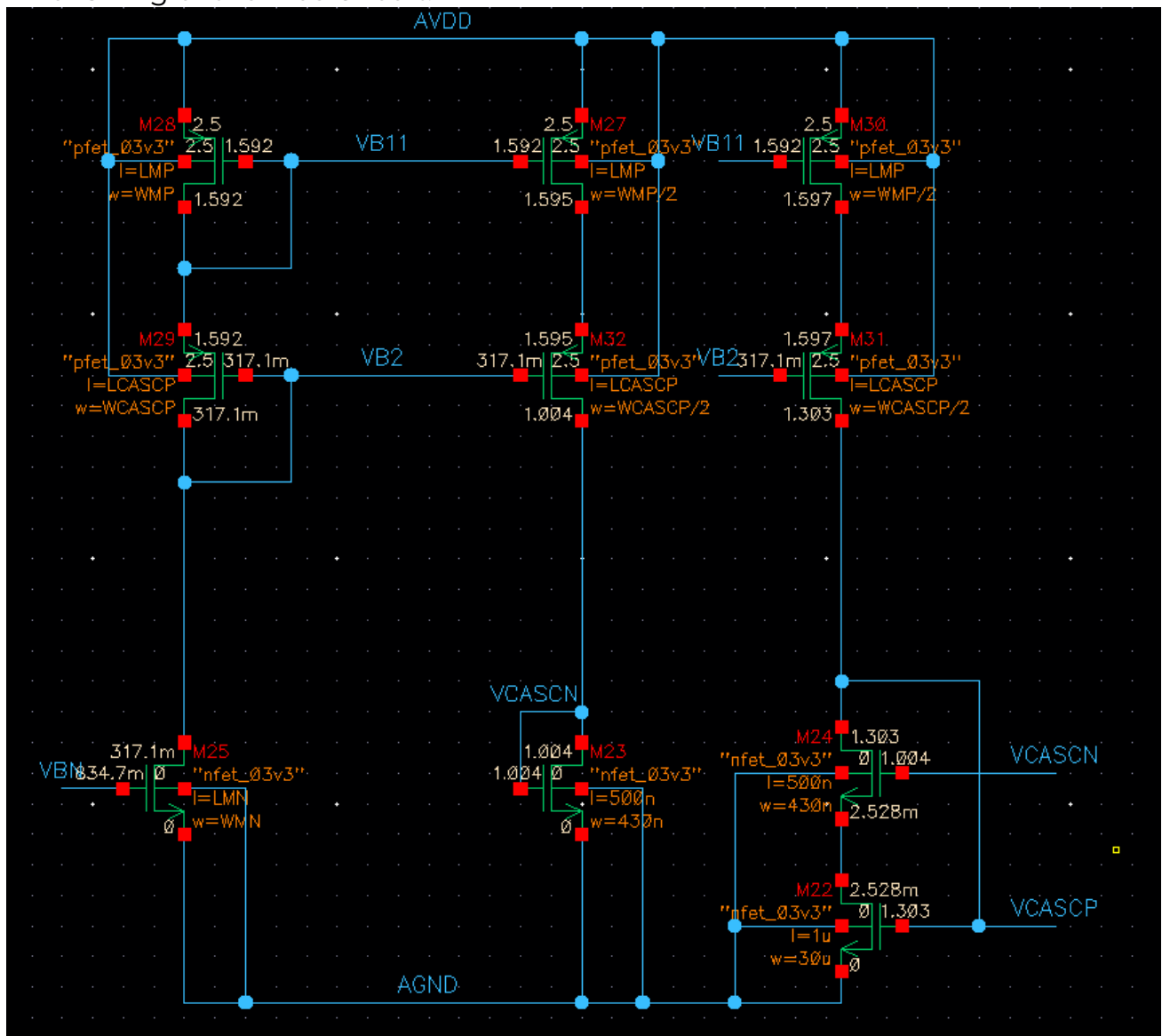


Figure 20 Final Biasing Circuit after fine tuning

I quickly realized I need more current to reach the required voltage so I adjusted the current mirror accordingly to then focused the rest of the time tuning on the bottom 3 transistor adjust their width accordingly to control their VGS and thus the values for VCASCN and VCASCP till they reach their values achieved in the previous section.

PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

Having finished designing and fine tuning the circuit on cadence virtuoso I decided to continue my simulations on Xschem and ngspice for the better control and automation they give me over the circuit.

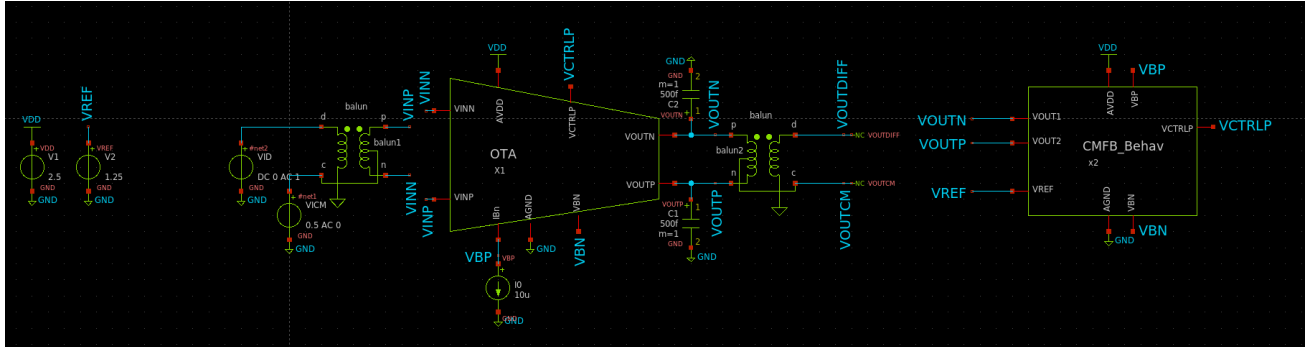


Figure 21 Xschem Testbench

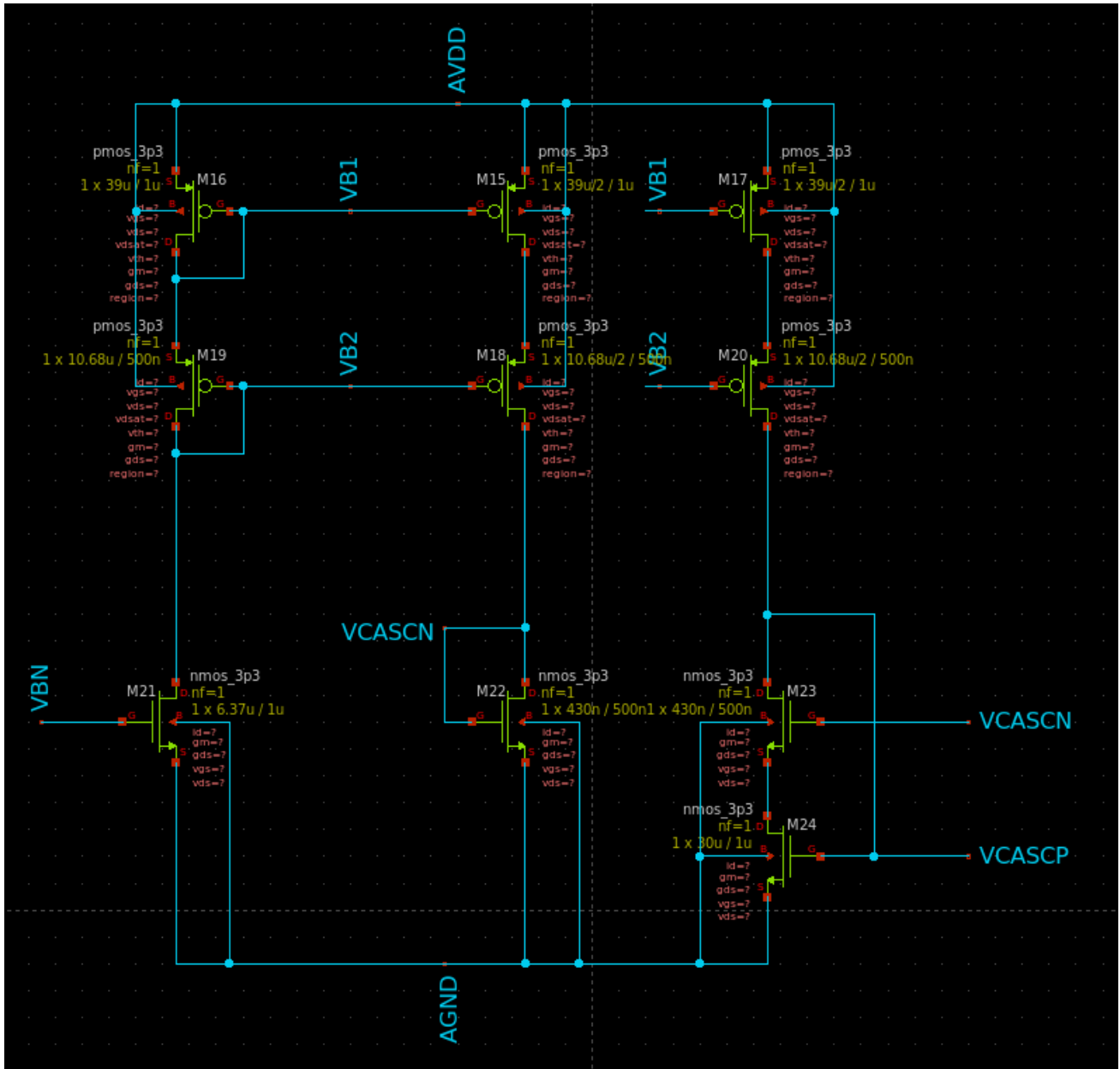


Figure 22 Biasing Circuit in Xschem

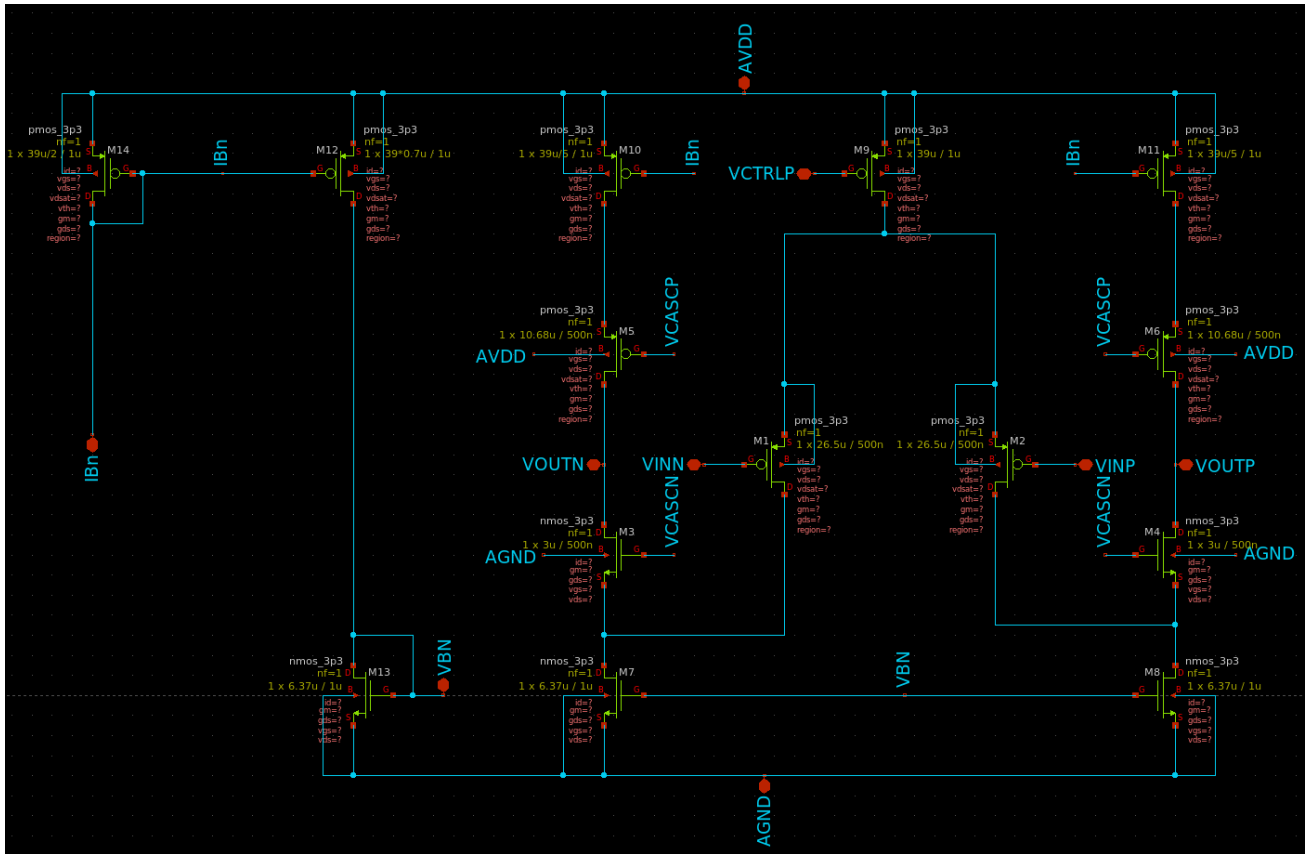


Figure 23 OTA in Xschem

OP Analysis:

Using VICM = 0.5V in the middle of CMIR

Using VREF at 1.25 in the middle of VDD for maximum output swing

```

save all

let x = 1
let num = 1
dowhile x <= 25
    set num = {${x}}
    save @m.x1.xm{${num}}.m0[id]
    save @m.x1.xm{${num}}.m0[vgs]
    save @m.x1.xm{${num}}.m0[vds]
    save @m.x1.xm{${num}}.m0[vdsat]
    save @m.x1.xm{${num}}.m0[vth]
    save @m.x1.xm{${num}}.m0[gm]
    save @m.x1.xm{${num}}.m0[gds]
    save @m.x1.xm{${num}}.m0[gmb]
    save @m.x1.xm{${num}}.m0[cgs]
    save @m.x1.xm{${num}}.m0[cgd]
    save @m.x1.xm{${num}}.m0[csb]
    save @m.x1.xm{${num}}.m0[cdb]
    let x = x + 1
end

*? OP Analysis
shell rm Testbench.raw
op
show m : id : gm : gmb : gds : vgs : vth : vds : vdsat
set appendwrite
write Testbench.raw

```

Figure 24 OP Analysis Code

To Showcase region in Xschem I used the TCL code provided by Amir Sameh on Canvas to show the region on the transistor symbol:

```

@model
nf=@nf
@m x @W / @L

tcleval(id=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(id)}]])
tcleval(vgs=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vgs)}]])
tcleval(vds=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vds)}]])
tcleval(vdsat=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vdsat)}]])
tcleval(vth=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vth)}]])
tcleval(gm=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(gm)}]])
tcleval(gds=[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(gds)}]])
tcleval(region=[

if {[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(id)}]] == "?" } {
    return "?"
}

if {[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(id)}]] < 6.3e-9 || [ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vgs)}]] < 1e-10 } {
    return "cutoff"
} else if {[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vgs)}]] < [ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vth)}]] } {
    return "subth."
} else if {[ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vds)}]] > [ngspice::get_node [subst -nocommand {v\@m.$(path)\@spiceprefix@name\m0(vdsat)}]] } {
    return "Sat."
} else {
    return "triode"
}
})

```

Figure 25 Region Symbol with region code

Results:

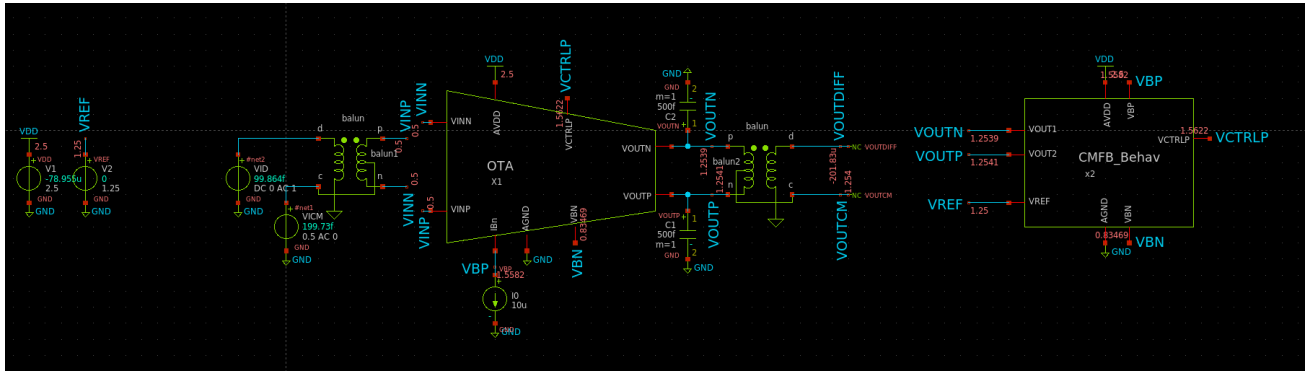


Figure 26 DC Voltages Annotated on Testbench Schematic

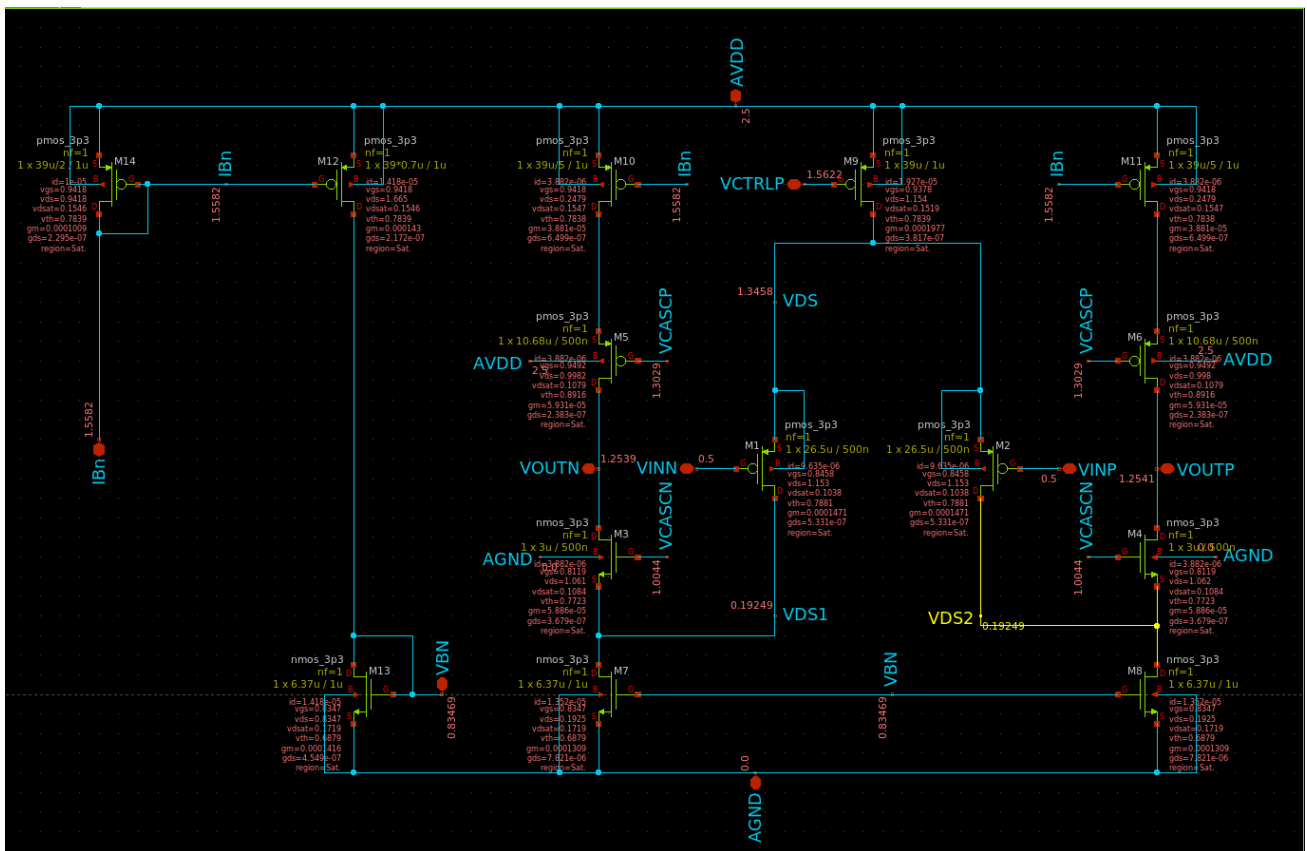


Figure 27 DC Voltages Annotated on OTA Schematic

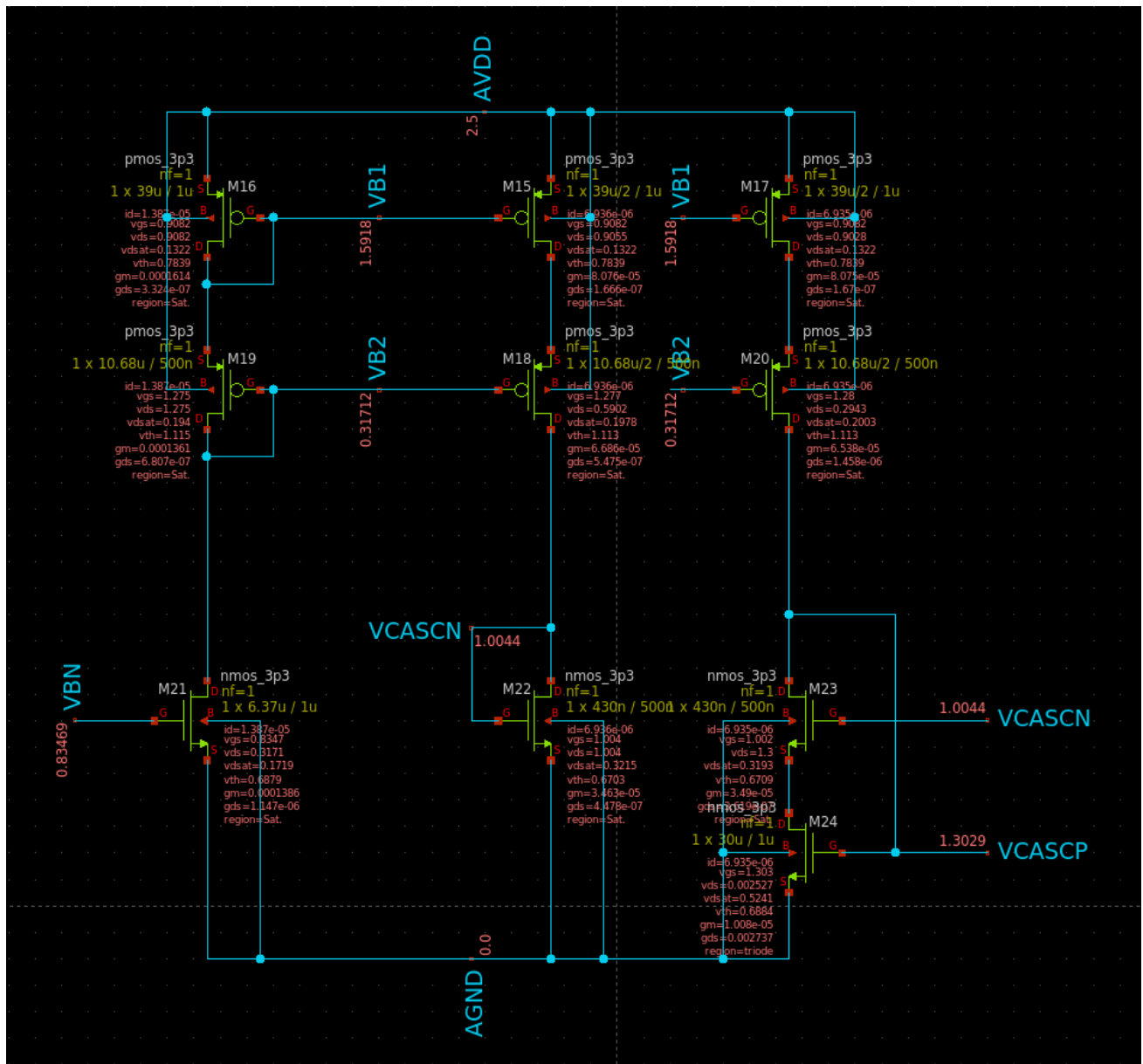


Figure 28 DC Voltages Annotated on Biasing Circuit

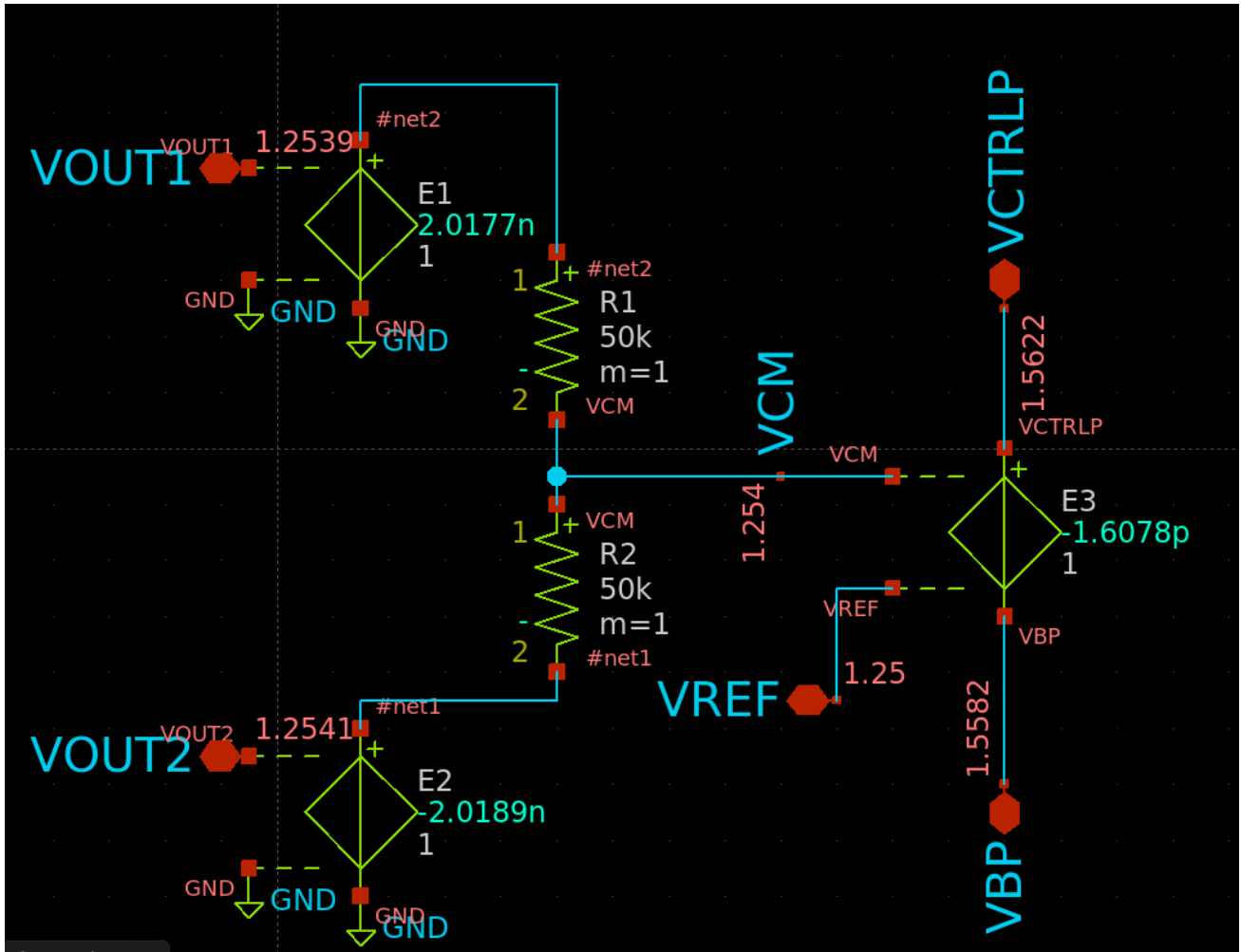


Figure 29 Ideal CMFB Voltages


```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm24.m0      m.x1.xm23.m0      m.x1.xm22.m0
model       nmos_3p3.13      nmos_3p3.0         nmos_3p3.0
id          6.93547e-06      6.93547e-06      6.93592e-06
vgs         1.30292         1.0018            1.00433
vds         0.00252739      1.30035           1.00431
vdsat       0.524148        0.319318          0.321513
vth         0.688439        0.670859          0.67033
gm          1.00787e-05      3.48959e-05      3.46337e-05
gds         0.00273702      3.61852e-07      4.47775e-07
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm20.m0      m.x1.xm18.m0      m.x1.xm21.m0
model       pmos_3p3.8      pmos_3p3.8         nmos_3p3.9
id          6.93548e-06      6.93592e-06      1.38656e-05
vgs         1.28008         1.27742           0.834687
vds         0.294284        0.590187          0.317116
vdsat       0.200315        0.197813          0.171902
vth         1.11262         1.113             0.687909
gm          6.53769e-05      6.68624e-05      0.000138635
gds         1.458e-06         5.47462e-07      1.14698e-06
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm13.m0      m.x1.xm7.m0        m.x1.xm8.m0
model       nmos_3p3.9      nmos_3p3.9         nmos_3p3.9
id          1.41839e-05      1.35172e-05      1.35172e-05
vgs         0.834687         0.834687          0.834687
vds         0.834684        0.192487          0.192487
vdsat       0.171903        0.171902          0.171902
vth         0.687908        0.687909          0.687909
gm          0.000141649        0.000130901      0.000130901
gds         4.5495e-07         7.82073e-06      7.82055e-06
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm3.m0        m.x1.xm4.m0        m.x1.xm10.m0
model       nmos_3p3.8      nmos_3p3.8         pmos_3p3.9
id          3.88233e-06      3.88234e-06      3.88235e-06
vgs         0.811858         0.811857          0.94182
vds         1.06145         1.06165           0.247856
vdsat       0.10844        0.108439          0.154745
vth         0.772303        0.772303          0.783798
gm          5.88604e-05      5.88605e-05      3.88109e-05
gds         3.67905e-07      3.67886e-07      6.49884e-07
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm11.m0      m.x1.xm17.m0      m.x1.xm16.m0
model       pmos_3p3.9      pmos_3p3.13        pmos_3p3.13
id          3.88235e-06      6.93548e-06      1.38656e-05
vgs         0.94182         0.90816           0.90816
vds         0.247855        0.902793          0.908159
vdsat       0.154745        0.132161          0.132161
vth         0.783798        0.783939          0.783939
gm          3.88109e-05      8.07521e-05      0.000161441
gds         6.4989e-07         1.66958e-07      3.32371e-07
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm15.m0      m.x1.xm14.m0      m.x1.xm12.m0
model       pmos_3p3.13      pmos_3p3.13        pmos_3p3.13
id          6.93592e-06      1e-05              1.41839e-05
vgs         0.90816         0.94182           0.94182
vds         0.905457        0.941819          1.66531
vdsat       0.132161        0.154634          0.154636
vth         0.783939        0.783939          0.783937
gm          8.0757e-05        0.000100902      0.000142976
gds         1.66618e-07      2.29461e-07      2.17158e-07
```

BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm9.m0	m.x1.xm19.m0	m.x1.xm5.m0
model	pmos_3p3.13	pmos_3p3.12	pmos_3p3.12
id	1.92698e-05	1.38656e-05	3.88234e-06
vgs	0.937771	1.27472	0.949224
vds	1.15416	1.27472	0.998194
vdsat	0.151877	0.193974	0.107913
vth	0.783939	1.11454	0.891561
gm	0.000197713	0.000136071	5.93079e-05
gds	3.81679e-07	6.80734e-07	2.38298e-07
BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm6.m0	m.x1.xm1.m0	m.x1.xm2.m0
model	pmos_3p3.12	pmos_3p3.12	pmos_3p3.12
id	3.88234e-06	9.6349e-06	9.6349e-06
vgs	0.949225	0.845839	0.845839
vds	0.997993	1.15335	1.15335
vdsat	0.107914	0.103818	0.103818
vth	0.891561	0.788091	0.788091
gm	5.93078e-05	0.000147097	0.000147097
gds	2.38321e-07	5.33147e-07	5.33147e-07

Figure 30 DC Op Point of all transistors

- What is the CM level at the OTA output?

$$V_{CMOUT} \approx 1.25 = V_{REF}$$

- What are the differential input and output voltages of the error amplifier? What is the relation between them?

$$VIN_{diff} = 1.254 - 1.25 = 0.004., \quad VOUT_{diff} = 1.5622 - 1.5582 = 0.004$$

The relationship between them is unity which is equal to the gain of error amplifier.

Differential Small Signal:

```
*? Small Signal
shell rm Testbench.raw
ac dec 10 1 10g

* DC Gain, BW, GBW and UGF Calculations
meas ac Gain MAX vmag(VOUTDIFF)
let Gain_dB = db(Gain)
let f3db_gain = Gain_dB - 3
meas ac BW WHEN vdb(VOUTDIFF) = f3db_gain
meas ac UGF WHEN vdb(VOUTDIFF) = 0
let GBW = Gain*BW

* PM Calculations
let voutdiff_ph = 180*cph(VOUTDIFF)/pi
meas ac phaseatzerogain find voutdiff_ph when vdb(VOUTDIFF)=0
let PM = 180 - abs(phaseatzerogain)

print Gain_dB
print GBW
print PM
plot xlog vdb(VOUTDIFF)
plot xlog voutdiff_ph
set appendwrite
write Testbench.raw
```

Figure 31 Diff SS Simulation Script

```
gain          = 3.555692e+03 at= 1.000000e+00
bw            = 1.159993e+04
ugf           = 4.130187e+07
phaseatzerogain = -9.764694e+01
gain_db = 7.101848e+01
gbw = 4.124578e+07
pm = 8.235306e+01
```

Figure 32 Simulation Results

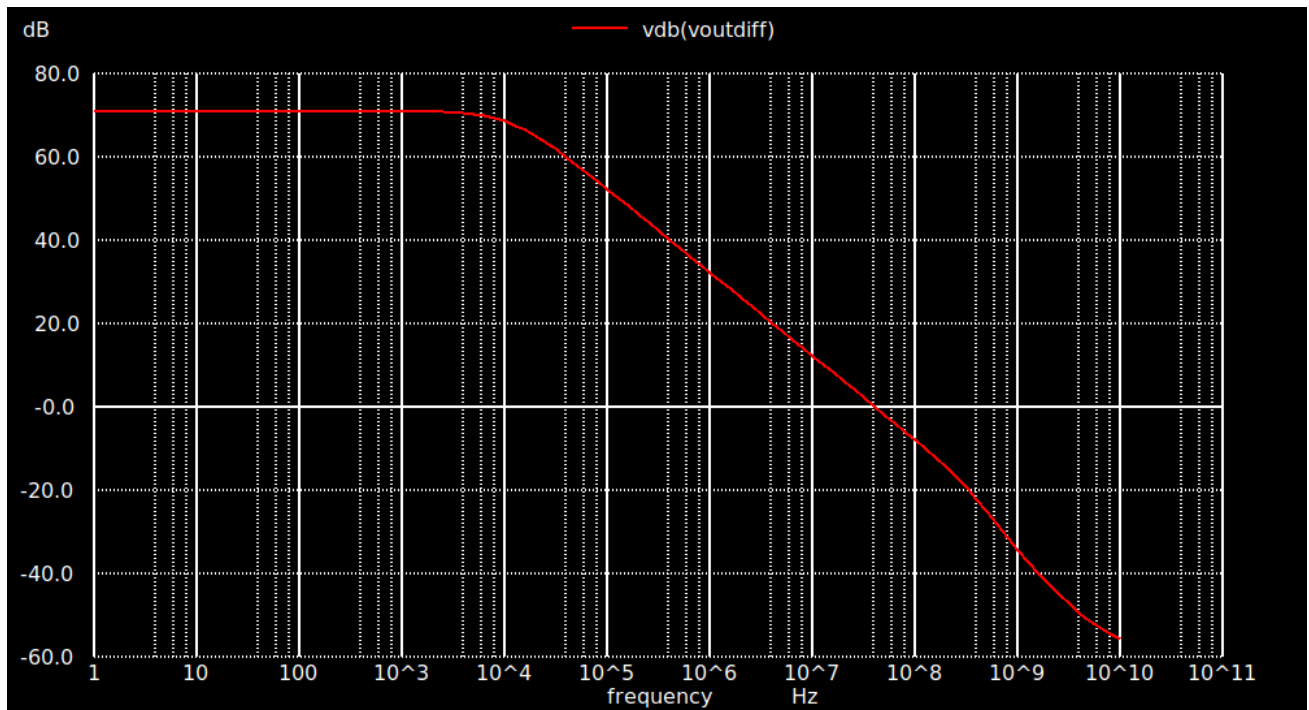


Figure 33 DC AOL Gain mag in dB

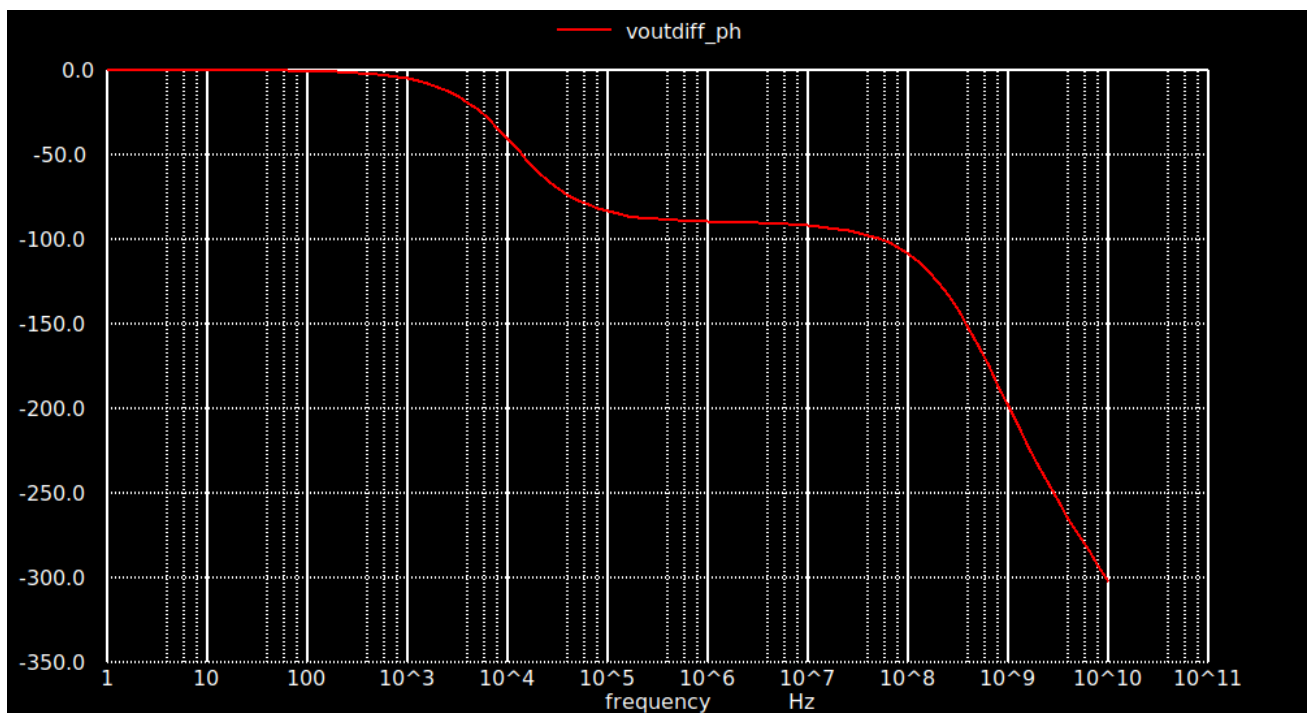


Figure 34 DC AOL Gain Phase vs Frequency

Hand Analysis:

$$R_{OUT} = [gm_{casen}ro_{casen}(r_{mn}/ro_{inp})]//gm_{cascp}ro_{cascp}ro_{mp} \approx \mathbf{18.3M\Omega}$$

$$A_{OL} = GmR_{OUT} = 147\mu * 18.3M = \mathbf{2.675K}$$

$$BW_{OL} = \frac{1}{2\pi R_{OUT}C_{OUT}} \approx \mathbf{17.5KHz}$$

$$GBW_{OL} = UGF = A_{OL} * BW_{OL} = \mathbf{46.812MHz}$$

	Simulation	Hand Analysis
Open Loop gain	3555.7	2675
BW	11.6KHz	17.5KHz
GBW	41.2MHz	46.812MHz
UGF	41.3MHz	46.812MHz

There is a difference between the simulation and analytical results possibly due to errors in calculating gds and due to neglecting the body effect of the cascode devices in the simulation resulting in a lower ROUT than expected thus a lower gain, but the results are still comparable to what's expected.

PART 4: Open-Loop OTA Simulation (Actual CMFB)

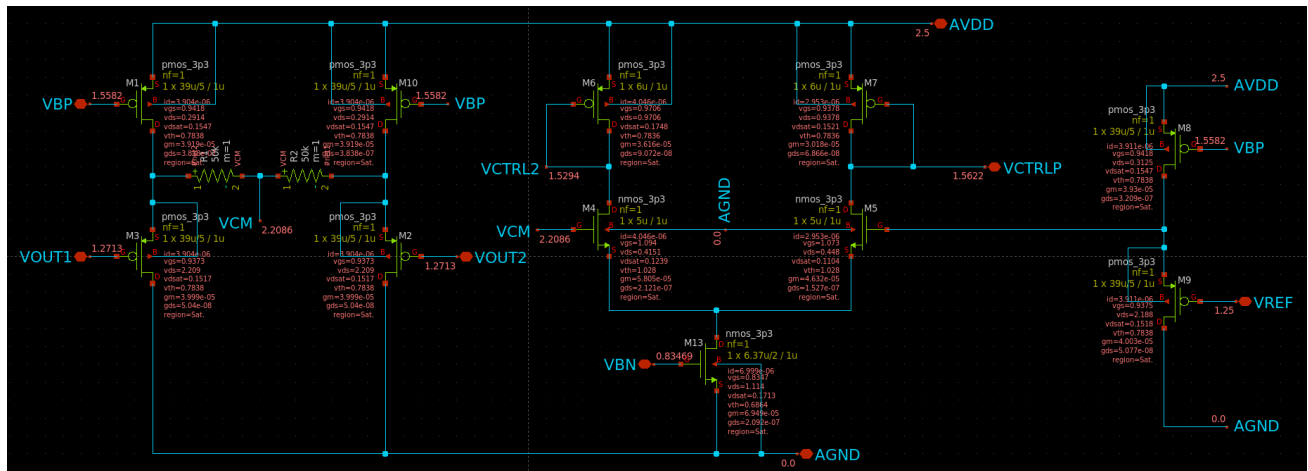


Figure 35 Real CMFB Network

Sizing Procedure:

Common Mode Sensing is done by a common drain, it is connected to the same mirror as the PMOS Mirror in the OTA and has the same sizing as them to run the same current.

For Simplicity I gave them all the same sizing as the Mirrors driving the current in the CG Branch.

Additionally I fed Vref through the same CD to account for the Voltage shift caused by $|V_{GSP}|$

This Leaves me with the differential amplifier pairs and their current mirror source:

The current mirror source is given the same sizing as the NMOS Sink Current sources in the OTA but with half the W/L for half the current budget of the OTA (**7uA**)

Error Amplifier:

All we have for the amplifier is that it requires a small gain nothing huge, so we'll assume unity gain or a bit higher than it

$$\text{let } A_v = \frac{g_{m_n}}{g_{m_p}} \approx 1.5$$

$$\text{Assume } \left(\frac{gm}{I_D} \right)_{in} = 15 \text{ (M)} , L = 1\mu m$$

$$I_D = 3.5\mu A$$

Into ADT with appropriate VDS:

LUT ?

Corner ☐ All ?

Temp (°C) ☐ All ?

Frequency ?

ID ?

gm/ID ?

L ?

VDS ?

VSB ?

itack ?

Results:

	Name	TT-27.0
1	ID	3.5u
2	IG	N/A
3	L	1u
4	W	5.06u
5	VGS	741.8m
6	VDS	833.3m
7	VSB	0
8	gm/ID	14.74

'GM = 51.57u'

$$L = 1\mu m , W = 5\mu m , g_{m_n} = 50\mu S$$

Accordingly, I can acquire the dimensions for the PMOS Load:

Using previous assumptions

$$gm_p = \frac{gm_n}{1.5} = 33.33\mu S$$

Assum $L = 1\mu m$
 $I_D = 3.5\mu A$

Into ADT with appropriate VDS:

LUT: pmos_03v3
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 3.5u
 gm: 33.33333u
 L: 1u
 VDS: 0.8333
 VSB: 0
 itack: 1

Results:

Name	TT-27.0
1 ID	3.5u
2 IG	N/A
3 L	1u
4 W	6.08u
5 VGS	954.7m
6 VDS	833.3m
7 VSB	0
8 gm/ID	9.462

Figure 36 CMFB PMOS Load Sizing

$$L = 1\mu m, W = 6\mu m$$

After a little bit of fine tuning, I increased W just slightly to better match Vref

$$L = 1\mu m, W = 7\mu m$$

Calculating new Output Swing:

$$\text{minimum Output Swing} = V_{CASCN}^* + V_{MN}^* \approx 0.3333$$

$$\text{maximum Output Swing} = V_{DD} - V_{GSP} - V_P^* = 2.5 - 0.9375 - 0.2 = 1.3625$$

$$\text{Middle of Output Swing} = V_{REF_{new}} = 0.847915 \approx \mathbf{0.85V}$$

Using these values for the output swing we can calculate values of R

$$R_1 + R_2 = \frac{V_{out_{MAX}} - V_{out_{MIN}}}{I_D} = 250K\Omega$$

$$R = 125K\Omega$$

Note: I added the correct caps towards the end of the Lab, the results didn't change much except in specific parameters at the end so as to save time and not repeat my analysis I kept all screenshots as is and only changed what changes with the current value of R.

Simulation:

I'll reuse the same codes as the previous part.

OP Analysis:

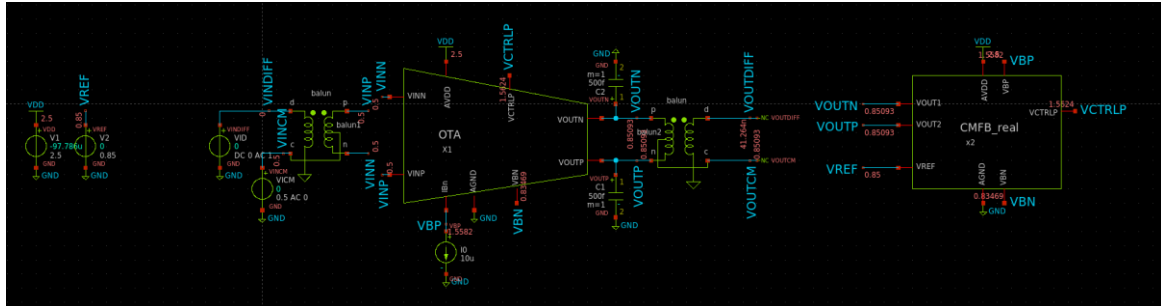


Figure 37 Testbench Voltages Annotated

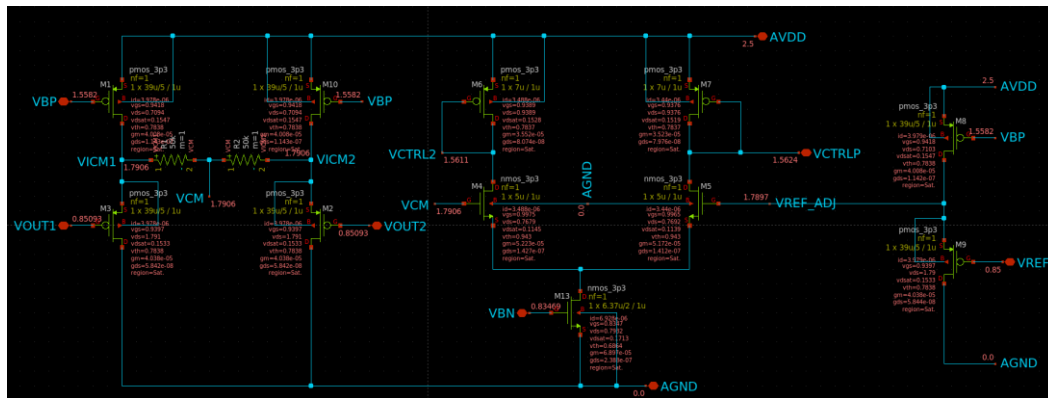


Figure 38 CMFB Voltages Annotated

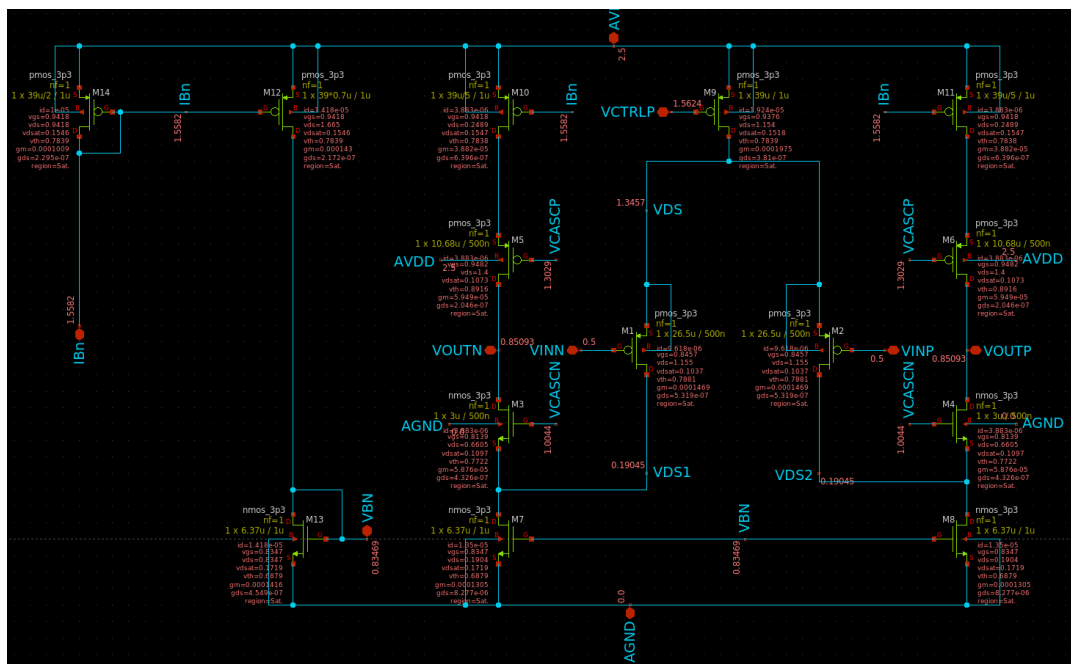


Figure 39 Latvia is actually Good at Winter

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm24.m0	m.x1.xm23.m0	m.x1.xm22.m0
model	nmos_3p3.13	nmos_3p3.0	nmos_3p3.0
id	6.93548e-06	6.93548e-06	6.93592e-06
vgs	1.30292	1.0018	1.00433
vds	0.00252739	1.30035	1.00431
vdsat	0.52415	0.319318	0.321513
vth	0.688439	0.670859	0.67033
gm	1.00786e-05	3.48959e-05	3.46337e-05
gds	0.00273703	3.61851e-07	4.47775e-07

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm20.m0	m.x1.xm18.m0	m.x2.xm13.m0
model	pmos_3p3.8	pmos_3p3.8	nmos_3p3.9
id	6.93548e-06	6.93592e-06	6.92759e-06
vgs	1.28008	1.27742	0.834687
vds	0.294283	0.590187	0.793168
vdsat	0.200315	0.197813	0.171345
vth	1.11262	1.113	0.686442
gm	6.53768e-05	6.68624e-05	6.89737e-05
gds	1.45802e-06	5.47462e-07	2.38831e-07

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x2.xm5.m0	m.x2.xm4.m0	m.x1.xm21.m0
model	nmos_3p3.9	nmos_3p3.9	nmos_3p3.9
id	3.43991e-06	3.48767e-06	1.38656e-05
vgs	0.996549	0.997471	0.834687
vds	0.769227	0.76788	0.317116
vdsat	0.113897	0.114477	0.171902
vth	0.942999	0.942999	0.687909
gm	5.1717e-05	5.22311e-05	0.000138635
gds	1.41183e-07	1.42722e-07	1.14698e-06

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm13.m0	m.x1.xm7.m0	m.x1.xm8.m0
model	nmos_3p3.9	nmos_3p3.9	nmos_3p3.9
id	1.41839e-05	1.35008e-05	1.35008e-05
vgs	0.834686	0.834687	0.834687
vds	0.834684	0.190444	0.190444
vdsat	0.171903	0.171902	0.171902
vth	0.687908	0.687909	0.687909
gm	0.000141649	0.00013046	0.00013046
gds	4.5495e-07	8.27737e-06	8.27737e-06

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm3.m0	m.x1.xm4.m0	m.x2.xm9.m0
model	nmos_3p3.8	nmos_3p3.8	pmos_3p3.9
id	3.88303e-06	3.88303e-06	3.9786e-06
vgs	0.813901	0.813901	0.939723
vds	0.660474	0.660474	1.78972
vdsat	0.109705	0.109705	0.15332
vth	0.772167	0.772167	0.783793
gm	5.87626e-05	5.87626e-05	4.03772e-05
gds	4.32623e-07	4.32623e-07	5.84402e-08

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x2.xm8.m0	m.x2.xm3.m0	m.x2.xm2.m0
model	pmos_3p3.9	pmos_3p3.9	pmos_3p3.9
id	3.9786e-06	3.97849e-06	3.97849e-06
vgs	0.94182	0.939719	0.939719
vds	0.710275	1.79064	1.79064
vdsat	0.154746	0.153317	0.153317
vth	0.783797	0.783793	0.783793
gm	4.00785e-05	4.03767e-05	4.03767e-05
gds	1.14195e-07	5.84192e-08	5.84192e-08

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x2.xm1.m0	m.x2.xm10.m0	m.x2.xm7.m0
model	pmos_3p3.9	pmos_3p3.9	pmos_3p3.9
id	3.97849e-06	3.97849e-06	3.43991e-06
vgs	0.94182	0.94182	0.937597
vds	0.709352	0.709352	0.937596
vdsat	0.154746	0.154746	0.151932
vth	0.783797	0.783797	0.783722
gm	4.00774e-05	4.00774e-05	3.52321e-05
gds	1.14313e-07	1.14313e-07	7.97648e-08

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x2.xm6.m0	m.x1.xm10.m0	m.x1.xm11.m0
model	pmos_3p3.9	pmos_3p3.9	pmos_3p3.9
id	3.48767e-06	3.88303e-06	3.88303e-06
vgs	0.938944	0.94182	0.94182
vds	0.938943	0.248917	0.248917
vdsat	0.152847	0.154745	0.154745
vth	0.783722	0.783798	0.783798
gm	3.55207e-05	3.88239e-05	3.88239e-05
gds	8.07417e-08	6.39584e-07	6.39584e-07

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm17.m0	m.x1.xm16.m0	m.x1.xm15.m0
model	pmos_3p3.13	pmos_3p3.13	pmos_3p3.13
id	6.93547e-06	1.38656e-05	6.93592e-06
vgs	0.90816	0.90816	0.90816
vds	0.902793	0.908159	0.905457
vdsat	0.132161	0.132161	0.132161
vth	0.783939	0.783939	0.783939
gm	8.0752e-05	0.000161441	8.07569e-05
gds	1.66958e-07	3.3237e-07	1.66617e-07

Figure 40 All Transistor OP Point

- **What is the CM level at the OTA output? Why?**

Voltage at the Output is equal to 0.85V, which is equal to Vref, because it is being set to VREF by the CMFB Loop

- **What are the differential input and output voltages of the error amplifier? What is the relation between them?**

$$\begin{aligned}
 VDIF_{IN} &= 1.7906 - 1.7897 = 0.0009 \\
 VDIF_{OUT} &= 1.5624 - 1.5611 = 0.0013 \\
 \frac{VDIF_{OUT}}{VDIF_{IN}} &= 1.4444
 \end{aligned}$$

The relationship between them is almost equal to the Small Signal gain the amplifier was designed to sustain while sizing it in the previous part.

Differential Small Signal:

Using the same setting as last time as well as same code

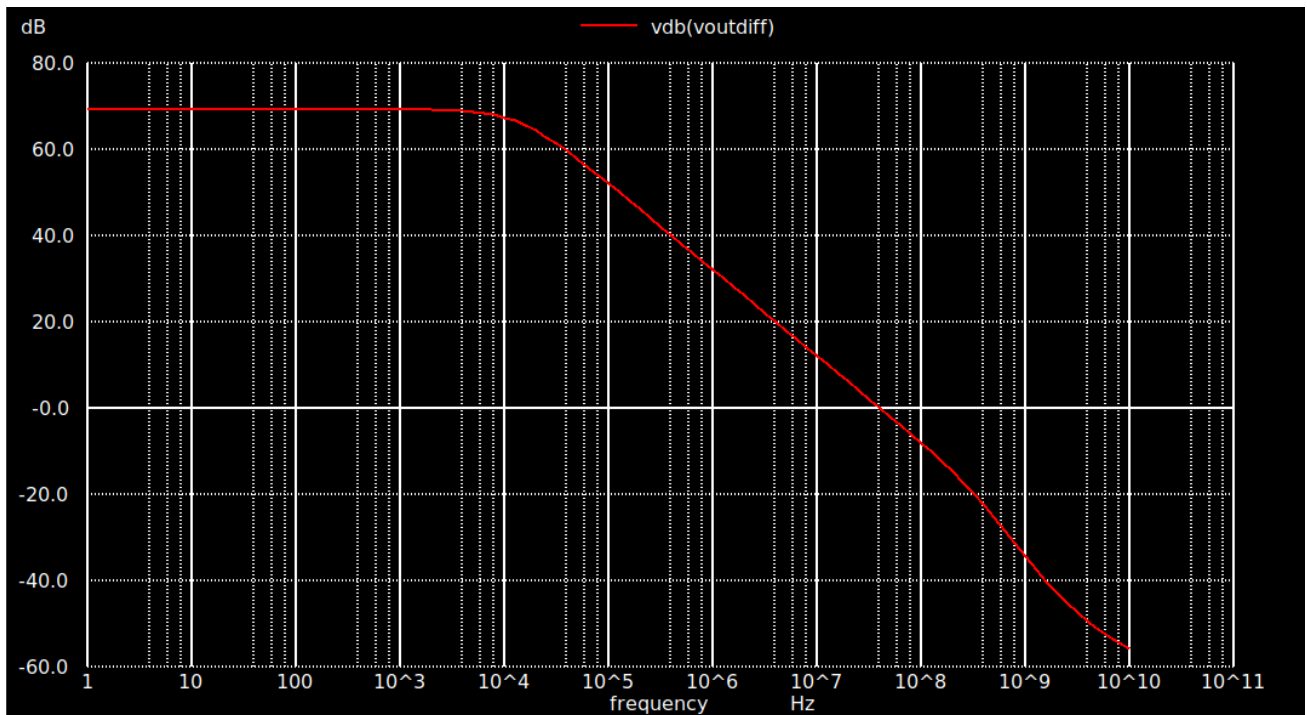


Figure 41 Diff Gain in dB

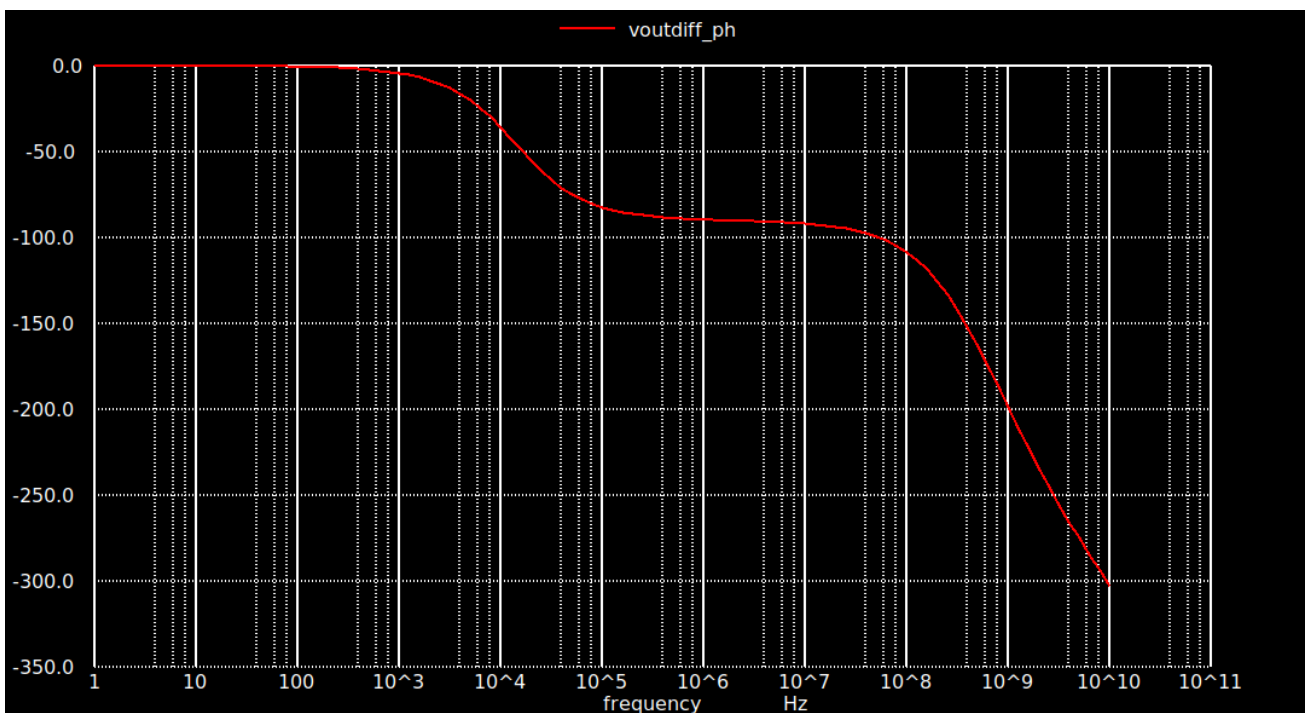


Figure 42 Diff Gain phase vs Freq

```

gain      = 2.907824e+03 at= 1.000000e+00
bw        = 1.383502e+04
ugf       = 4.018701e+07
phaseatzerogain = -9.744349e+01
gain_db = 6.927136e+01
gbw       = 4.022980e+07
pm        = 8.255651e+01

```

Figure 43 Circuit Parameters

	Behavioral	Real
Open Loop gain	3555.7	2907.8
BW	11.6KHz	13.83KHz
GBW	41.2MHz	40.2298MHz
UGF	41.3MHz	40.187MHz
PM	82.3	82.5565

There is a slight decrease in gain from the Behavioral to the real and a slight increase in BW, meaning the Output resistance possibly decreased due to changing the output cm voltage

I fixed this by increasing the bias voltages a little bit by increasing the current running through the Bias Circuit.

```

gain      = 3.847100e+03 at= 1.000000e+00
bw        = 1.239826e+04
ugf       = 4.753487e+07
phaseatzerogain = -1.005942e+02
gain_db = 7.170267e+01
gbw       = 4.769735e+07
pm        = 7.940580e+01

```

Figure 44 After Tweaking

PART 5: Closed Loop Simulation (AC and STB Analysis)

OP Analysis:

Done using the same code as before but only adding an extra save loop to save the component parameters of the CMFB network:

```

let x = 1
let num = 1
dowhile x <= 26
  set num = {$&x}
  save @m.x2.xm{$num}.m0[id]
  save @m.x2.xm{$num}.m0[vgs]
  save @m.x2.xm{$num}.m0[vds]
  save @m.x2.xm{$num}.m0[vdsat]
  save @m.x2.xm{$num}.m0[vth]
  save @m.x2.xm{$num}.m0[gm]
  save @m.x2.xm{$num}.m0[gds]
  save @m.x2.xm{$num}.m0[gmb]
  save @m.x2.xm{$num}.m0[cgs]
  save @m.x2.xm{$num}.m0[cgd]
  save @m.x2.xm{$num}.m0[csb]
  save @m.x2.xm{$num}.m0[cdb]
  let x = x + 1
end

```

Figure 45 Extra Save dowhile loop for CMFB

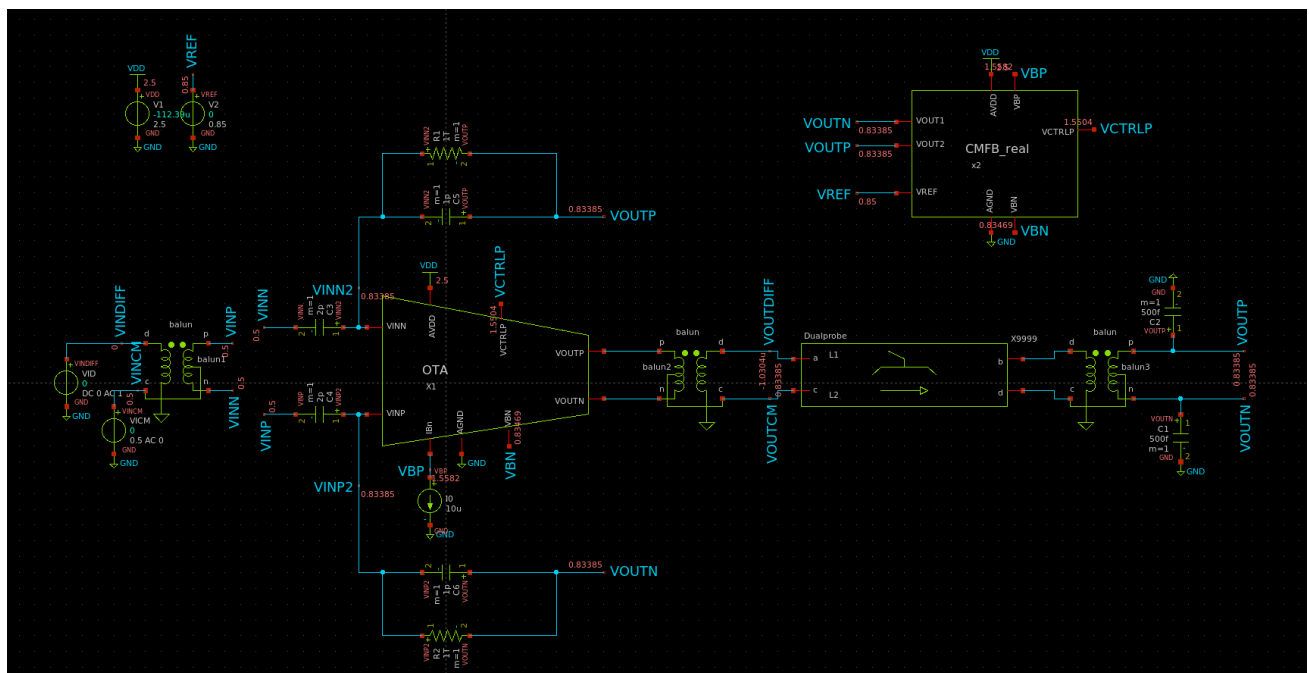


Figure 46 DC Voltages of Testbench Annotated

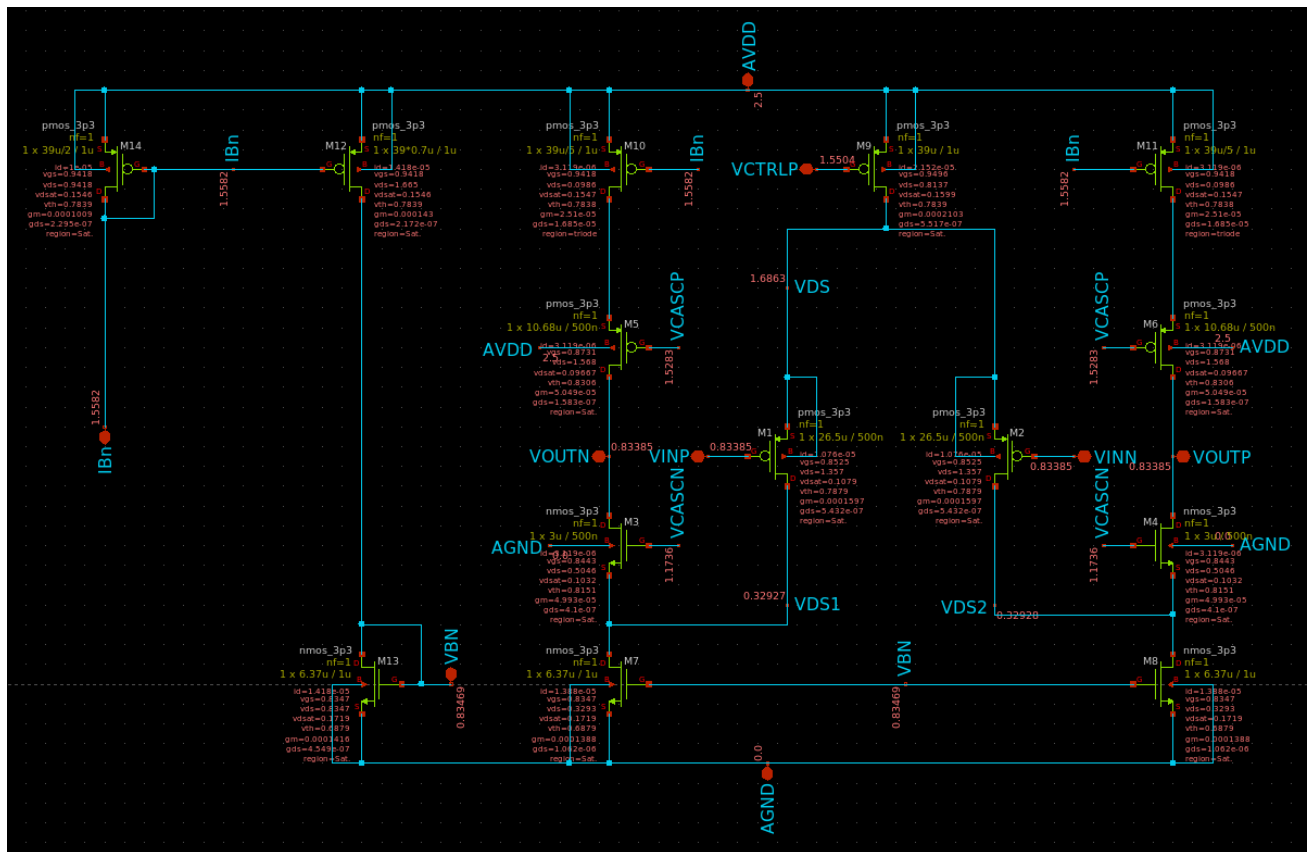


Figure 47 OP Point and DC Voltages of OTA

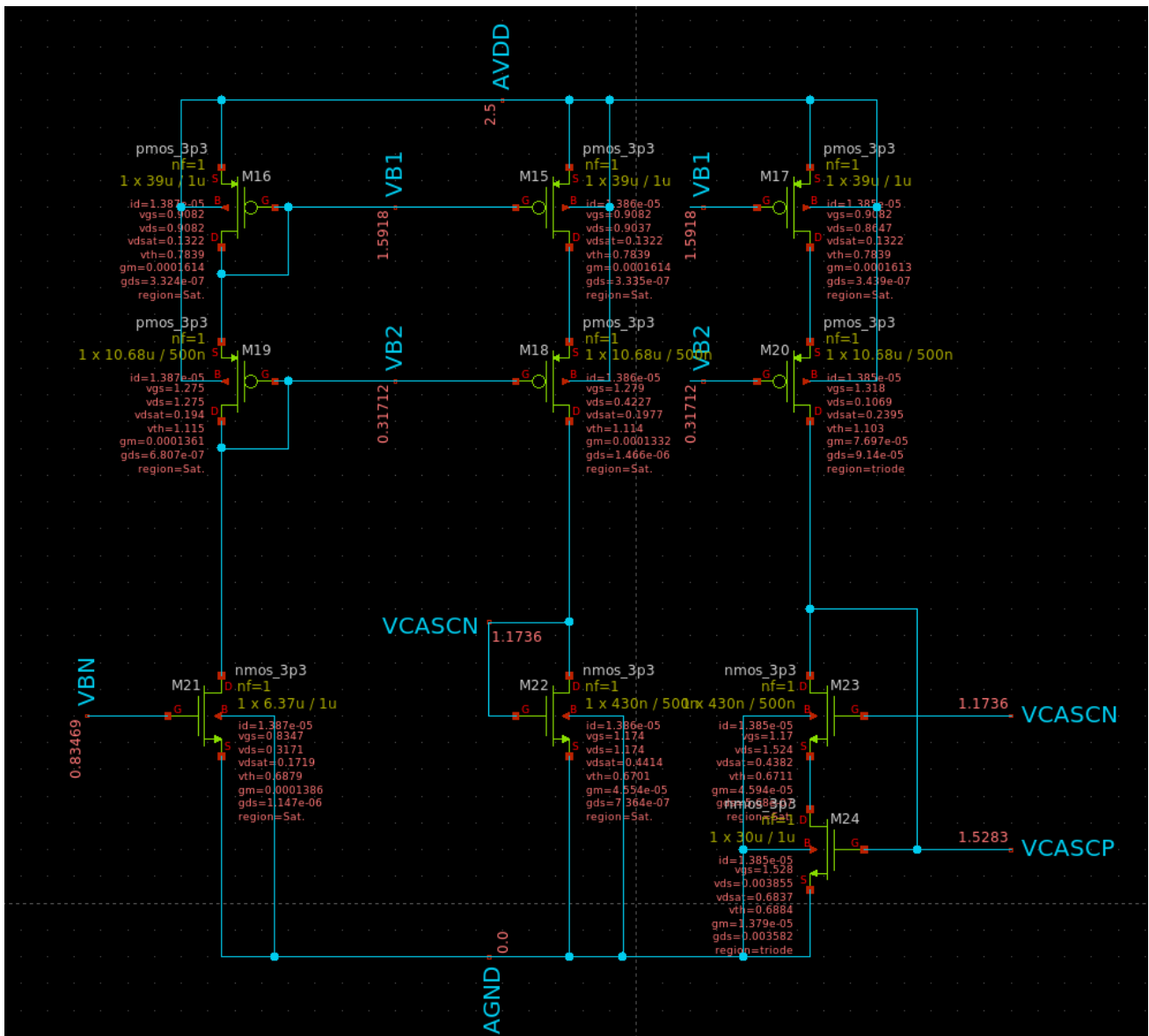


Figure 48 OP Point and DC Voltages of Bias Circuit

As seen here the current is higher than before in the bias circuit as I increased it by increasing the mirror ratio of the upper transistors to increase the bias voltages and increase the Open Loop gain of the OTA

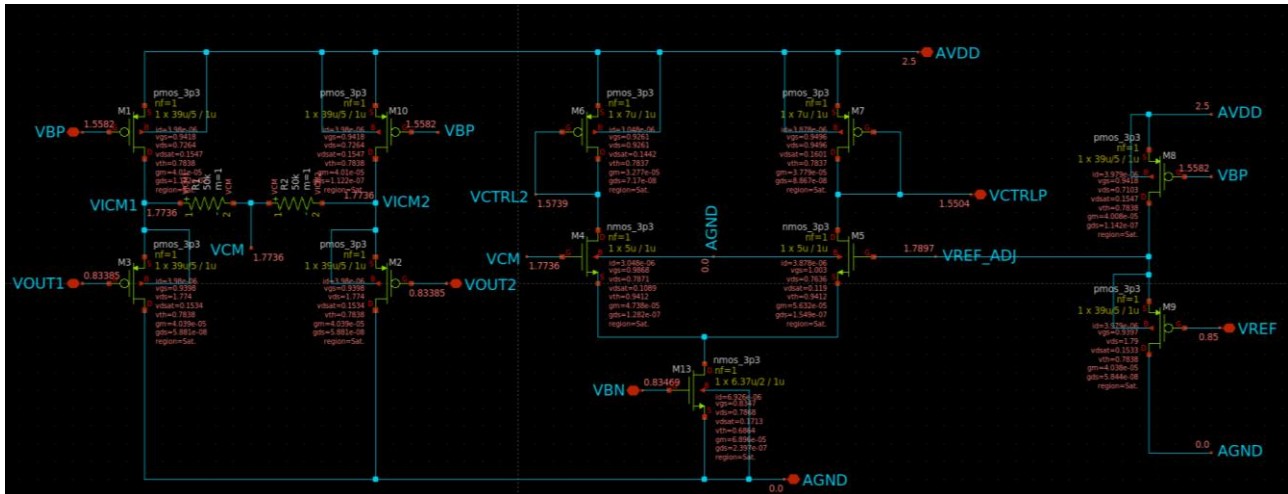


Figure 49 CMFB OP Point annotated and DC Voltages as well

- **What is the CM level at the OTA output? Why?**

It is equal to 0.83V almost equal to 0.85V the Reference voltage which is it being set to by the CMFB.

- **What is the CM level at the OTA input? Why?**

It is also equal to 0.83V as it is being set to the same voltage as the output by feedback

Differential Closed-Loop Response:

Again, using the same code as the previous parts for AC analysis and measures for the circuit parameters requested.

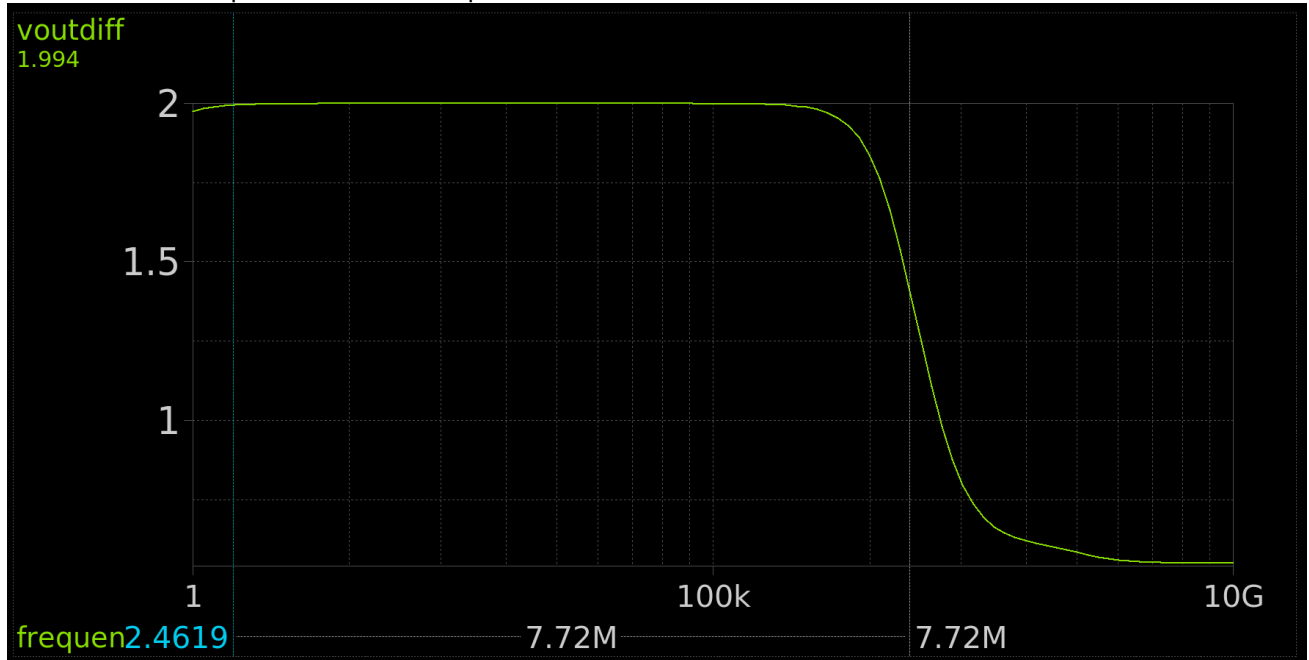


Figure 50 VOUTIFF vs Freq, CL Gain and CL BW Annotated

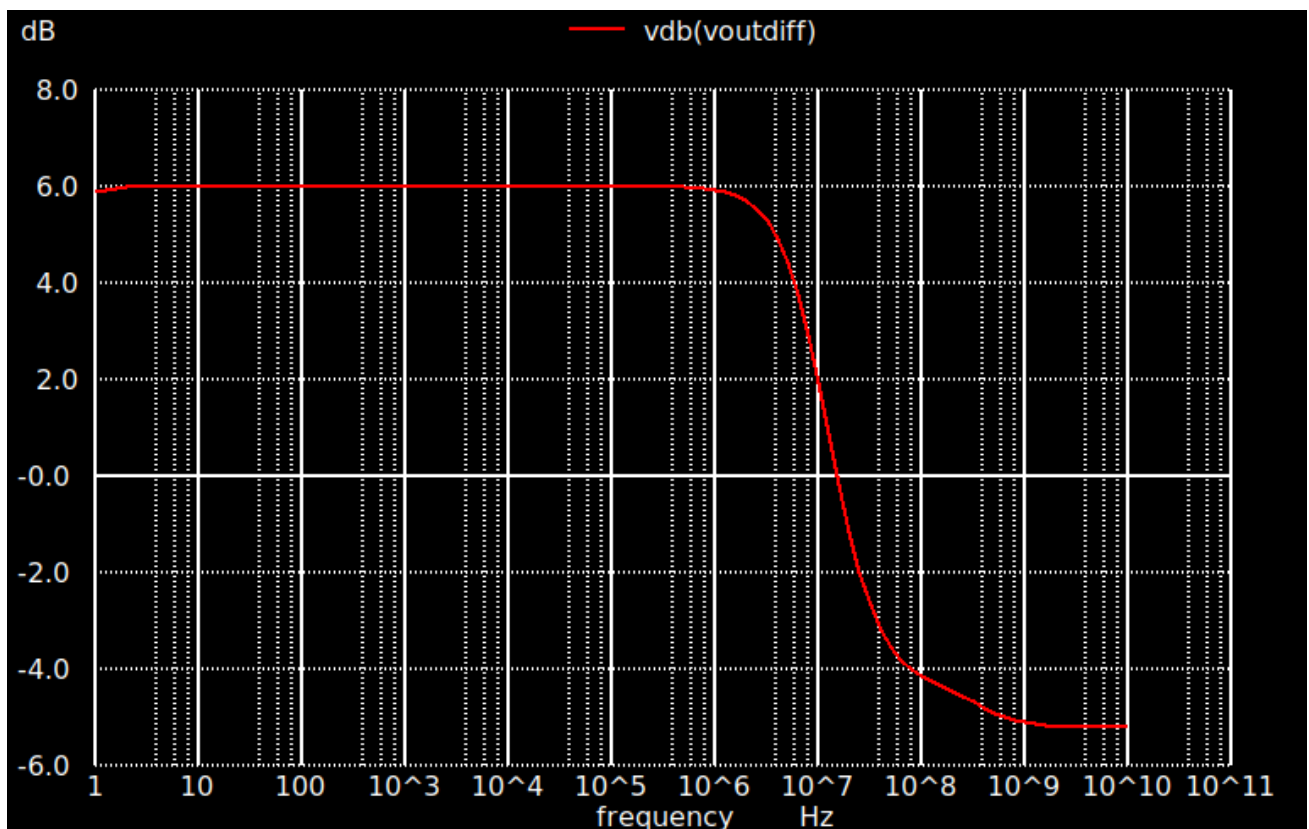


Figure 51 VOUTDIF in dB vs Freq

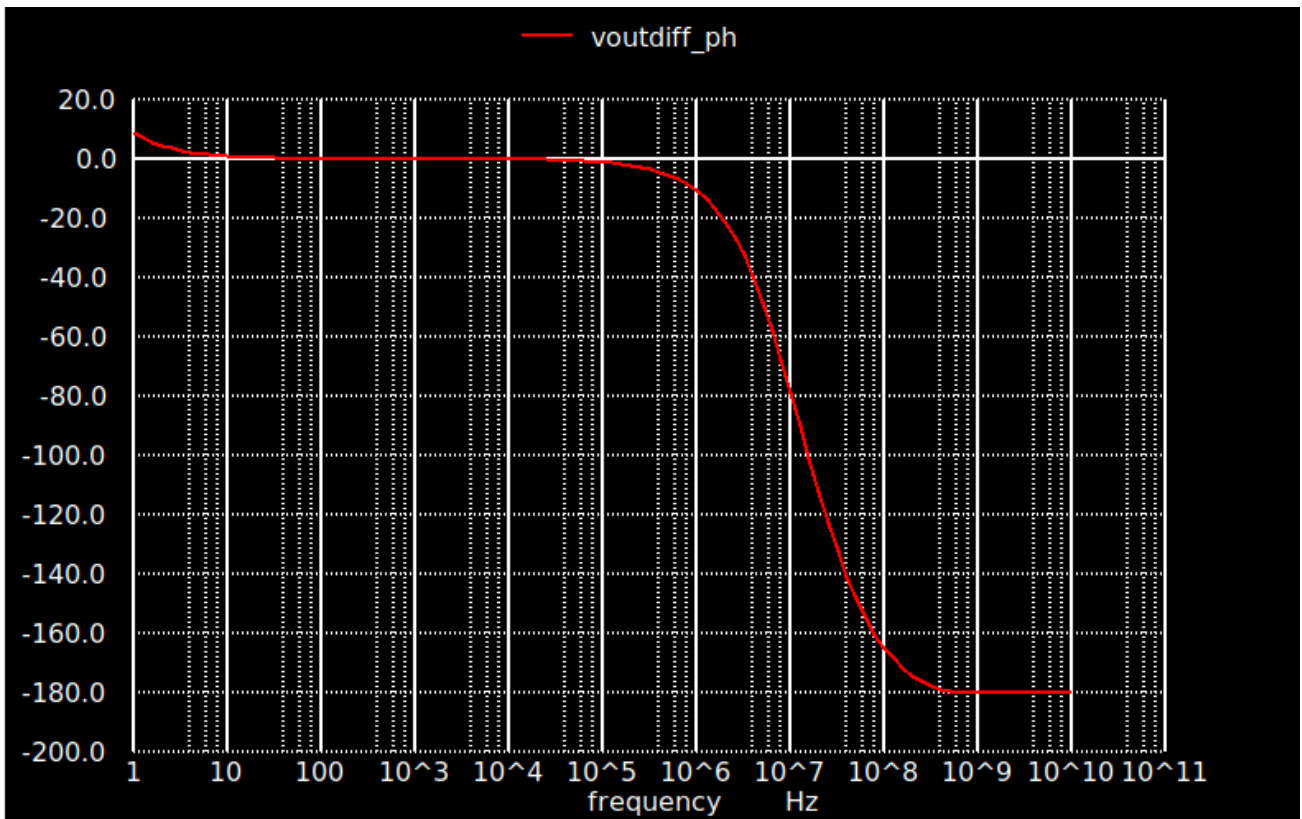


Figure 52 VOUTDIFF Phase vs Freq

```

gain          = 1.998389e+00 at= 1.000000e+03
bw            = 7.729439e+06
ugf           = 1.533007e+07
phaseatzerogain = -9.953494e+01
gain_db       = 6.013601e+00
gbw           = 1.544643e+07

```

Figure 53 Circuit Parameters Measured in Simulation

	Simulation
DC Gain	1.9983
BW CL	7.7294MHz
GBW CL	15.4464MHz

STB Analysis:

For STB Analysis I used the DualProbe provided by Amir Sameh on Canvas with its code in it



Figure 54 Loop Gain for CM and Diff Feedbacks

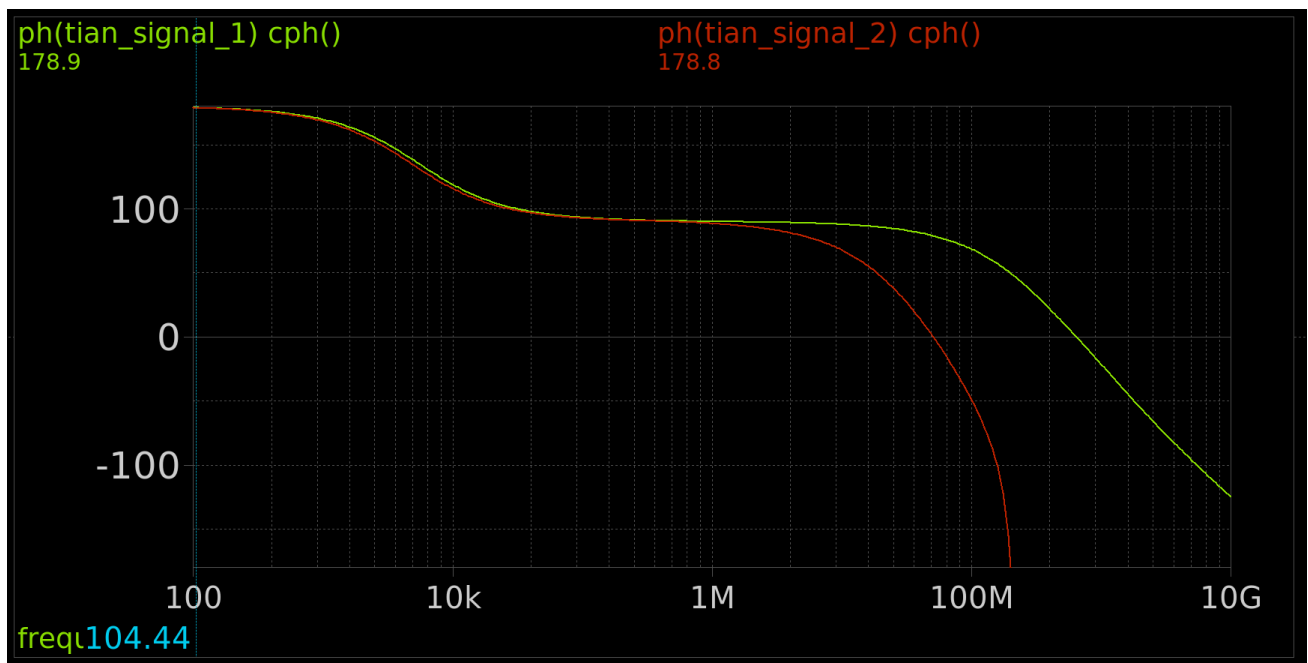


Figure 55 Phase for CM and Diff Feedbacks

```

loop1_lg = 6.178642e+01
loop1_gx = 6.801794e+06
loop1_pm = 8.854762e+01
-----
loop2_lg = 6.561742e+01
loop2_gx = 9.104580e+06
loop2_pm = 6.971373e+01

```

Figure 56 Circuit Parameters obtained from Simulation

	Diff Loop	CM Loop
LG	1216.18	1905.46
LG dB	61.7	65.6
GBW	6.8MHz	9.1045MHz
PM	88.547	69.713

- **Compare GBW and PM of diff and CM loops. Comment.**

Differential has less GBW but higher PM compared to CM, Making CM less stable

- **Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation. Comment**

The AOL gain in the Open Loop simulation was equal to 3847.1 which gives a LG third that equal to 1282.36 which is close to the value we get, It is a bit different due to the loading effect of the feedback that we don't see in open loop, we tried to account for it though in the open loop simulation when designing the amplifier.

Due to the lower gain we also get lower GBW

PART 6: Closed Loop Simulation (Transient Analysis)

```
*? Transient Analysis

shell rm Testbench_cl.raw
save all

tran 10n 3u

remzerovec
write Testbench_cl.raw
```

Figure 57 Transient Analysis Code

Differential and CMFB loops stability (transient analysis) + CL settling time: Differential input pulse:

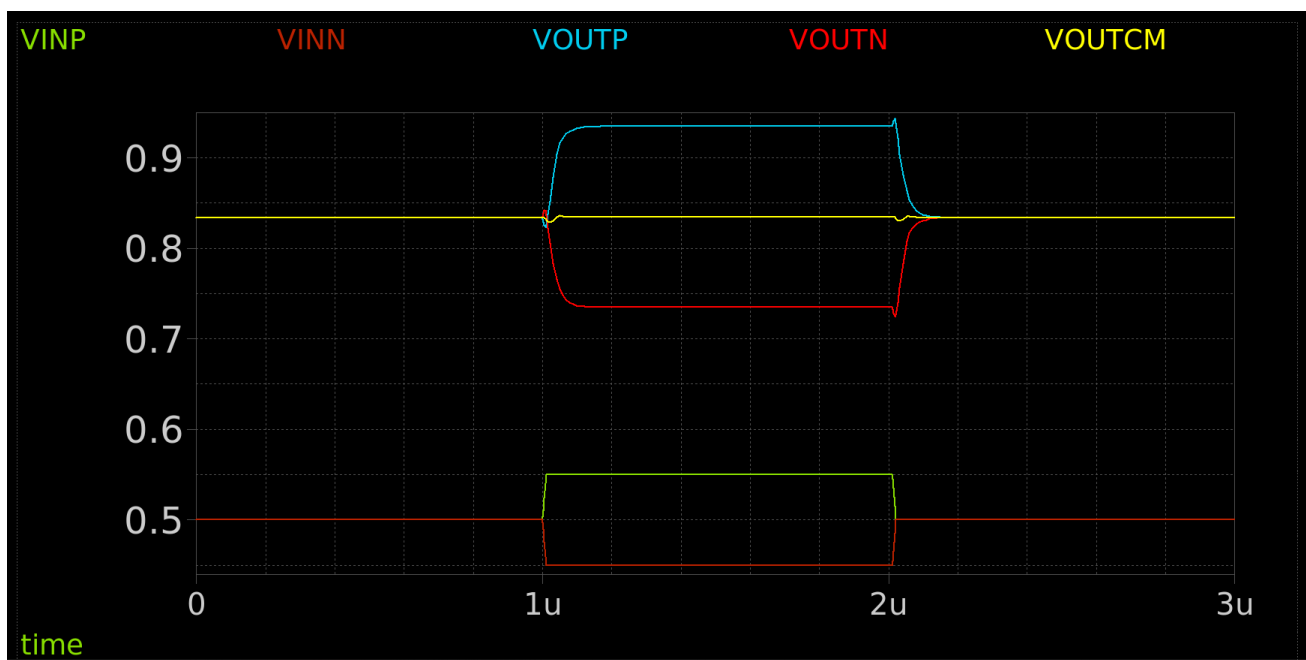


Figure 58 VINP VINN VOUTP VOUTN VOUTCM Overlaid

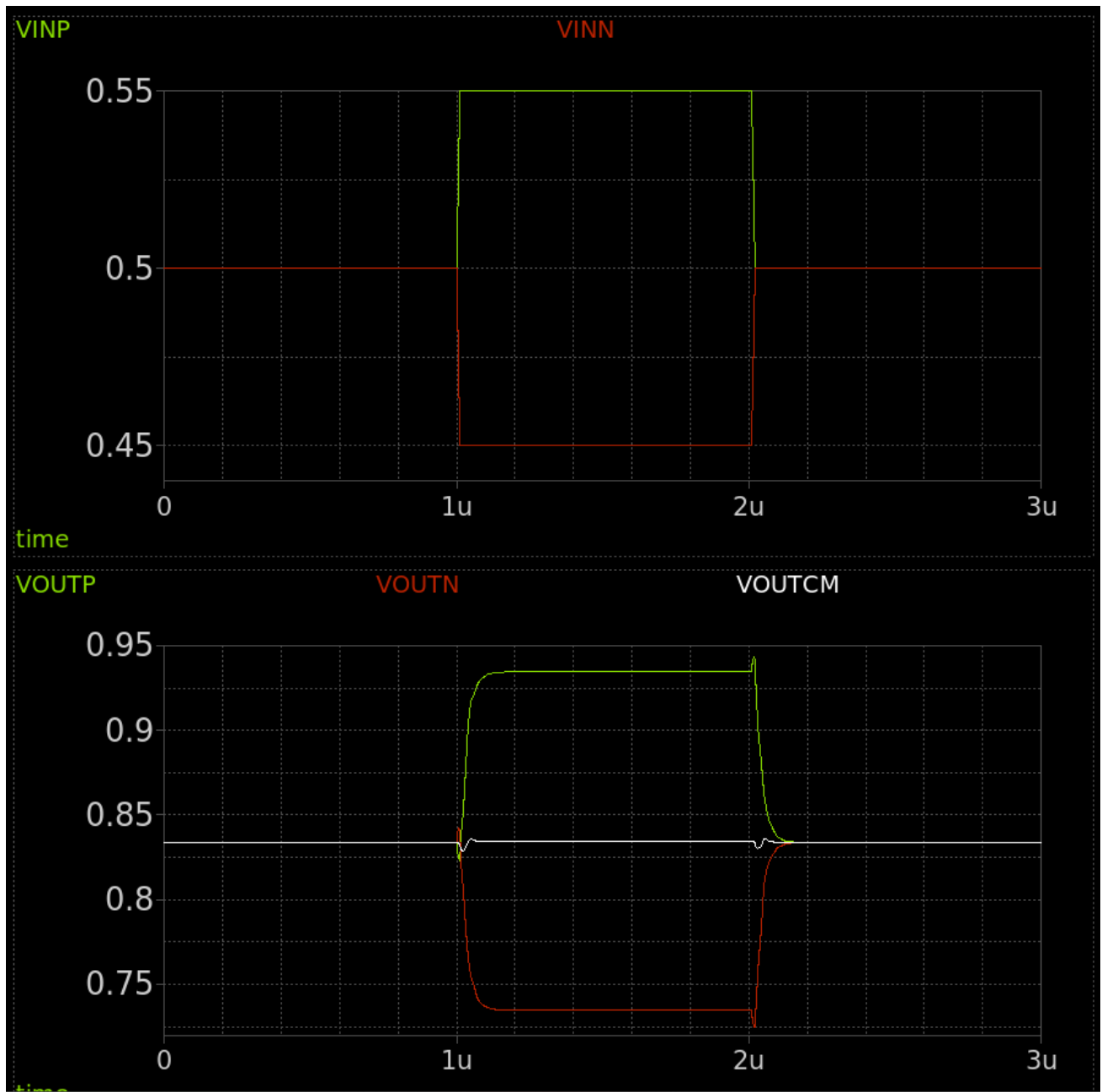


Figure 59 Inputs and Outputs Overlaid on Different Figures

- **Do you notice any differential/CM ringing? Are both loops stable with adequate PM?**

There is a little bit of barely visible ringing in CM due to its lower PM and none in Differential.

The spikes seen are caused by the capacitors in the circuit not allowing sudden changes.

Both have adequate PM that is within spec and doesn't distort the signal too much.

- **Calculate the 1% settling time and compare it to the required specification. If the specification is not satisfied, what design changes could be a possible solution?**

```
tran 10n 3u
meas tran FINALVAL FIND V(VOUTDIFF) AT=1.5u
meas tran INITIALVAL FIND V(VOUTDIFF) AT=0.5u
let SETTLEVAL = 0.99 * FINALVAL

meas tran t_settle TRIG V(voutdiff) AT = 1u TARG V(voutdiff) VAL = SETTLEVAL RISE=1
```

Figure 60 Code to Measure the Settling Time

I set the code to Trigger at 1u since this is when the signal starts getting injected into the amplifier as per the pulse.

```
finalval      = 1.998397e-01
initialval    = -3.683729e-06
t_settle      = 1.160286e-07 targ= 1.116029e-06 trig= 1.000000e-06
```

Figure 61 Settle time from Simulation

Settle time is equal to 116ns which is slightly above spec.

It can be improved by improving the GBW which can be improved by increasing gain slightly (Increasing W of input Pair to increase its gm) or increasing current to increase BW

I decided to test by increasing the Gain, I increased the widths of the input pair as well as the lengths of the cascode devices.

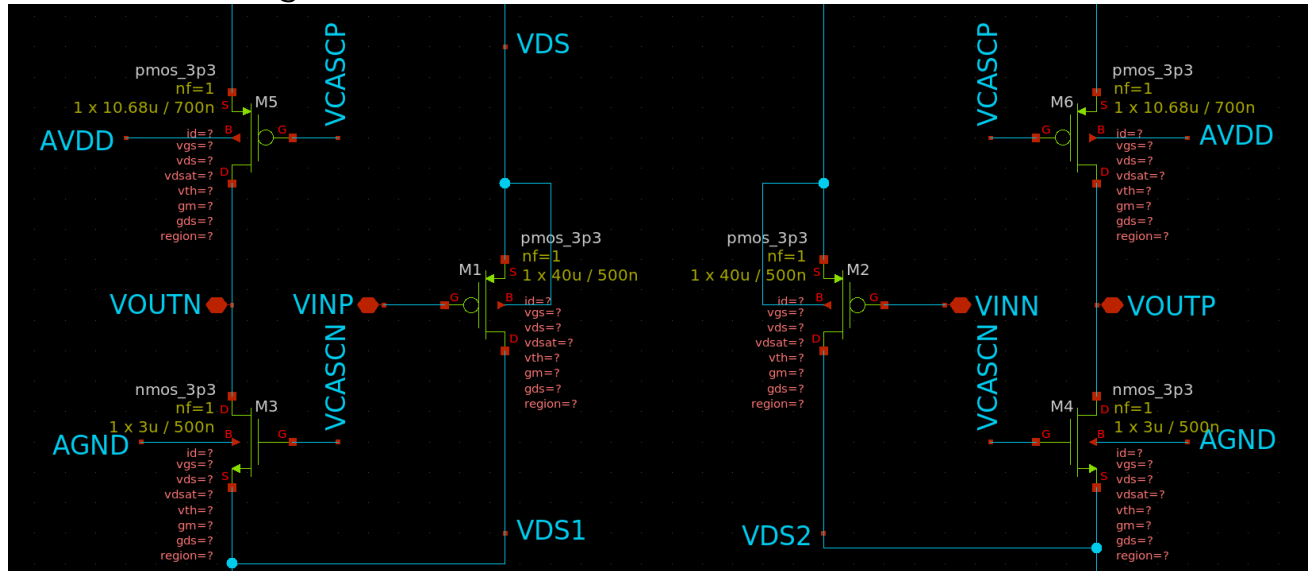


Figure 62 Adjusted Sizing for increased Settling time

```
finalval      = 1.998904e-01
initialval    = 9.946795e-07
t_settle      = 1.008043e-07 targ= 1.100804e-06 trig= 1.000000e-06
```

Figure 63 Improved Settling Time

Differential and CMFB loops stability (transient analysis): CM input pulse:

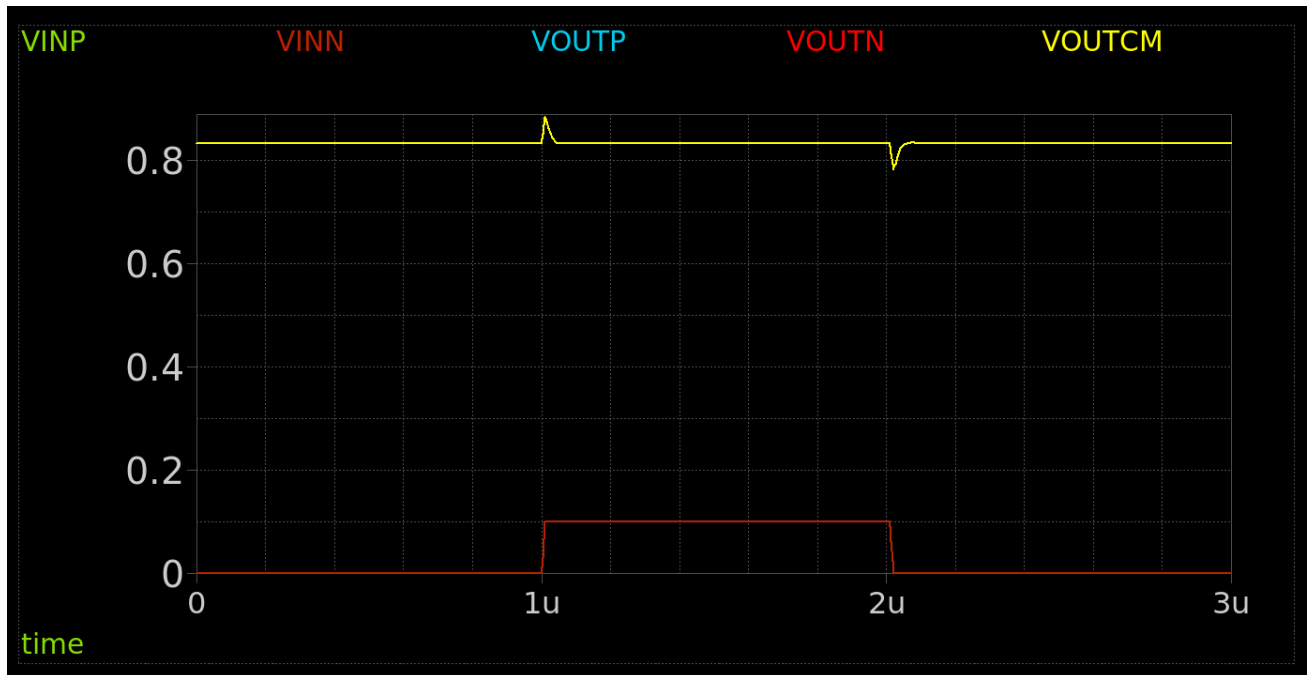


Figure 64 VINP VINN VOUTP VOUTN VOUTCM Overlaid on same figure

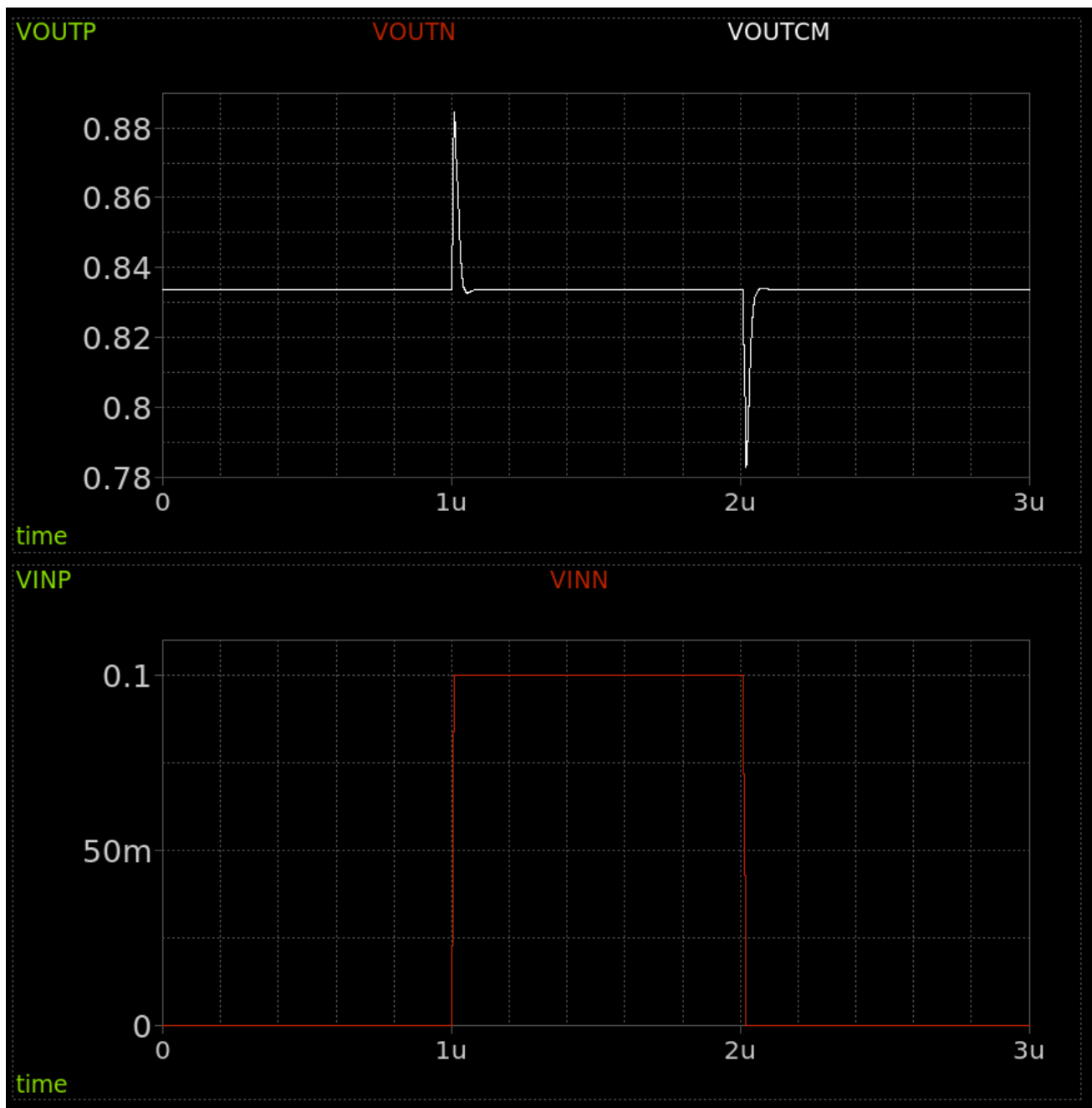


Figure 65 Inputs and Outputs Overlaid on Different Figures

Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

There is a little bit of CM ringing barely visible other than that it seems decently stable.

The spikes seen are due to the capacitors not allowing sudden changes.

Both have adequate PM that is within spec and doesn't distort the signal too much.

Output Swing:

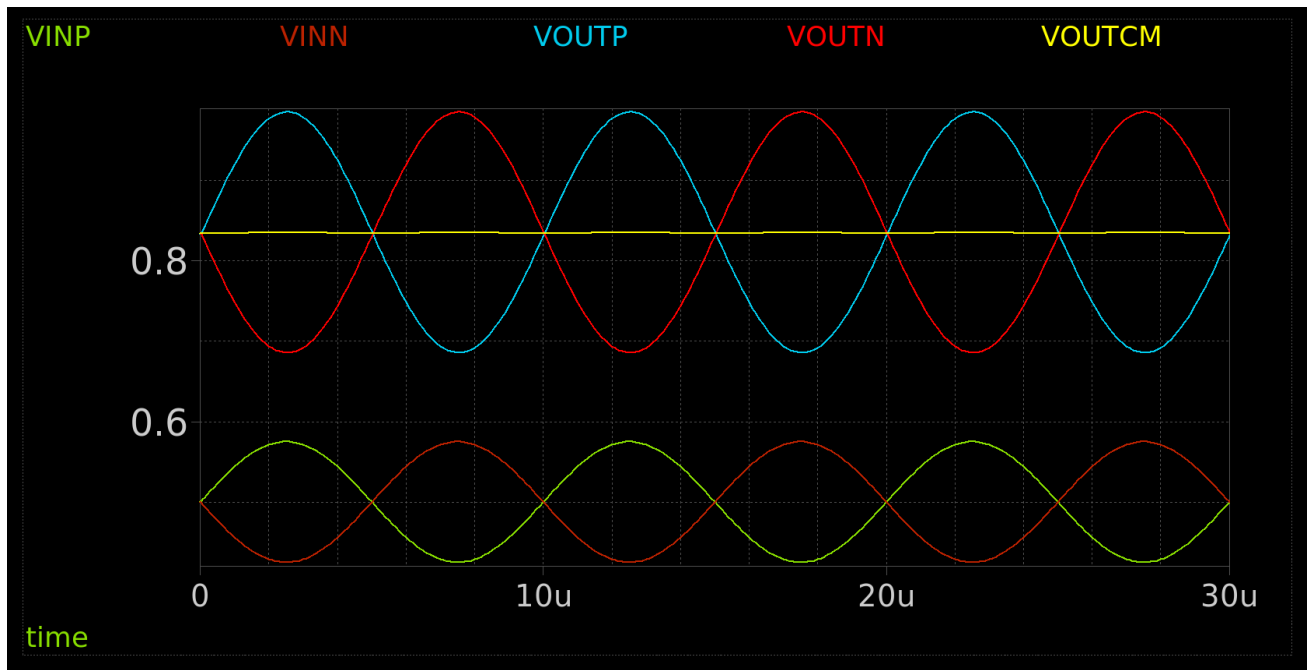


Figure 66 VINP VINN VOUTP VOUTN VOUTCM Overlaid

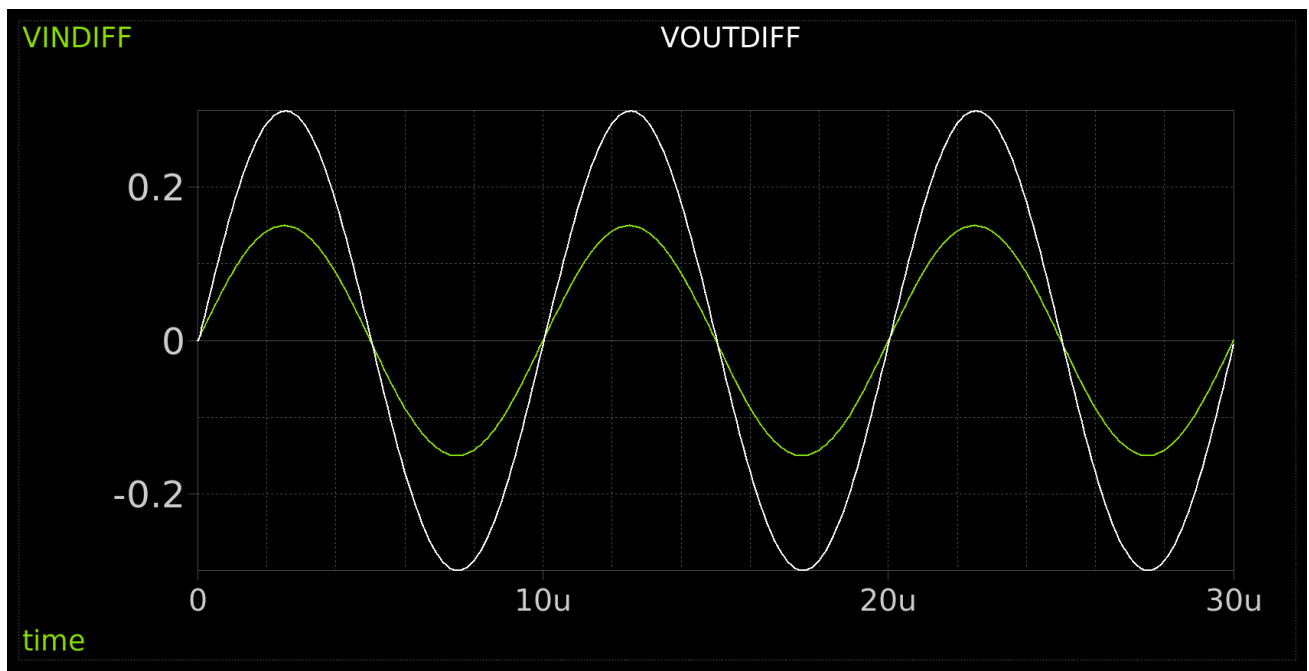


Figure 67 VINDIFF and VOUTDIFF Overlaid

```
vpp_diff_in      = 2.999993e-01
vpp_diff_out     = 5.993310e-01
dc_gain = 1.997775e+00
```

Figure 68 PeakToPeak Values and DC Gain

```
* Sinuosoidal
tran 0.1u 30u
meas tran VPP_DIFF_IN PP V(VINDIFF)
meas tran VPP_DIFF_OUT PP V(VOUTDIFF)

let DC_GAIN = VPP_DIFF_OUT / VPP_DIFF_IN
print DC_GAIN
```

Figure 69 Code for the sinusoidal signal